

Mobile Application Development: PetCare

Milestone 1 + Milestone 2 + Milestone 3

Propose

Assoc. Prof. Twittie Senivongse

By

Danuwat Maoleethong	6772037021
Patiphon Puntusin	6772055221
Paknathuch Wattanamongkolchoke	6772085021
Arissara Tirapaiboon	6772110021
Panarin Padungkij	6772074121
Techasit Srirueng	6870109221
Jakapong Chomklin	6870030221

This document is an integral component of the subject

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Executive Summary

PetCare Co., Ltd., founded in 2025 in Bangkok, is a PetTech startup dedicated to transforming pet healthcare and lifestyle management in Thailand. With the country's pet market expanding at a CAGR of nearly 9% through 2032, driven by rising numbers of "pet parents," there is an urgent need for integrated, intelligent digital solutions that go beyond fragmented record-keeping and manual care practices.

The PetCare Application is proposed as a **unified mobile platform** that consolidates health records, care reminders, safety tracking, nutrition guidance, and digital certification into one secure, AI-powered ecosystem. It addresses six core pain points faced by pet owners to day Health Records, Care Routines, Pet Safety, Nutrition, Wellness Tracking

The project runs from **1 September 2025 to 13 November 2026** (≈ 14.5 months for core delivery). nder an **Agile Scrum** methodology with **4-week sprints**, structured into **three delivery phases**:

- **Phase 1 – Core Foundation (Sep 2025 – Feb 2026):** Focus on building core functions such as digital health records, reminders, and pet profiles to attract and onboard pet owners.
- **Phase 2 – Partnership & Ecosystem Expansion (Mar – Jul 2026):** Strengthen platform credibility through collaborations with clinics, vets, and pet product partners.
- **Phase 3 – Global Expansion & Advanced Intelligence (Aug – Nov 2026):** Scale the platform internationally and integrate advanced AI features for predictive wellness and community engagement.

The development team is organized into four groups: Management/Product, Development & Innovation, Marketing & Customer Service, and Operations.

History of Document Change

Date	Detail	Change	Version
1/09/68	Project Proposal Created for Milestone 1	-	1.0
2/10/68	Add Milestone 2 and revised Milestone 1	- Add Scope Baseline - Add Schedule Baseline - Revised Milestone 1	2.0
3/11/68	Add Milestone 3 and revised Milestone 1+2	- Revised Milestone 1 + 2 - Add Milestone 3	3.0

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1 Business Case with financial analysis and exhibit

1.1 Introduction and Background



Figure 1 PetCare Co., Ltd., commercialize trademark

PetCare Co., Ltd., established in 2025 and headquartered in Bangkok, Thailand, is a PetTech startup dedicated to developing a mobile application that enhances pet healthcare and lifestyle management through innovation. Its flagship product, the Digital PetCare Assistant, empowers pet owners to manage health records, receive reminders for essential activities, track pet safety with IoT-enabled devices such as smart collars and GPS trackers, and access AI-driven nutritional and health insights. The application also generates digital health certificates that can be used as trusted references for breeding, trading, or consultations.

Our Vision to be Thailand's leading innovator in AI-driven digital pet care, enhancing the wellbeing of pets and their families while setting new standards for healthy, safe, and comprehensive pet management that honors pets as cherished family members.

Our Mission is to create accessible and reliable mobile solutions that leverage artificial intelligence to support pet owners in managing health and daily care with confidence. By offering proactive recommendations, predictive reminders, and personalized insights, PetCare seeks to transform pet care into a data-driven and preventive practice that enhances the overall well-being of pets.

1.2 Project Objective

1. Launch an integrated mobile application that centralizes pet health records, reminders, and safety tracking.
2. Provide nutrition logging and guidance with curated rules and human-reviewed content.
3. Offer a wellness dashboard with actionable insights for owners and clinics.
4. Enable integrations via APIs for clinics and vetted partners (e.g., IoT devices, insurers).

1.3 Project Background and As-is System

1.3.1. Project Background

Thailand's pet market is expanding rapidly—its pet food segment is projected to grow at ~8.8% CAGR (2026–2032), according to Verified Market Research. This growth is driven by the increasing number of "pet parents" who view their pets as family members and are willing to invest in their health and well-being. Despite this market growth, digital pet care solutions in Thailand remain underdeveloped and fragmented. Most existing tools provide only basic record-keeping or single-purpose features, leaving pet owners without a comprehensive, intelligent system for proactive care.

The responsibility of pet ownership extends beyond companionship to encompass careful planning and continuous care. This project is grounded in the recognition that pet owners universally pursue five key objectives for their animals' well-being

- Love: Promoting physical activity, emotional well-being, and a sense of contentment.
- Safety: Ensuring the security, traceability, and physical protection of pets within their environment.
- Food: Providing high-quality, growth-supporting, and long-term sustenance.
- Health: Maintaining a regimen of preventive care, including vaccinations, routine medical examinations, and timely treatment of ailments.
- Intelligent Support: Utilizing contemporary tools, such as data-driven recommendation systems and artificial intelligence, to deliver personalized insights and proactive guidance.

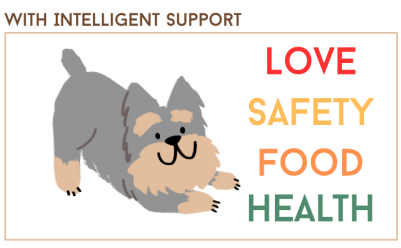


Figure 2 Four pillars of pet care with intelligent support

Pet care remains challenging due to the fragmented nature of relevant information. Critical data is often dispersed across disparate formats and locations, including paper records for medical history, digital calendars for reminders, product labels for nutritional information, and independent smart devices for activity monitoring. **This lack of data integration and centralization** creates significant friction, hindering owners' ability to efficiently and holistically manage their pets' care.

The proposed application is designed to directly address this critical gap. By centralizing disparate pet-related data and converting it into actionable insights, the system will not only empower owners to meet the fundamental goals of pet care but also generate valuable aggregated data. This data can, in turn, be leveraged to develop innovation across the pet care ecosystem, enabling stakeholders such as pet food manufacturers, veterinary clinics, and insurance providers to develop more effective, needs-based products and services. The platform's business model will be built upon this aggregated, anonymized data, which will be licensed to industry partners to inform product development, marketing strategies, and service improvements.

1.3.2. As-is System

Currently, pet care activities are managed through fragmented tools and manual processes. Pet owners rely on paper records, separate mobile applications, and ad-hoc information sources. The following overview describes the As-is System, highlighting the key actors, processes, and limitations of the current practice.

1.3.2.1. Actors

1. Owner (Pet Parent)

   Responsible for managing pet health, care routines, and daily activities using fragmented tools.

2. Veterinary Clinic

   Provides medical services, records vaccinations, and issues health certificates.

3. Third-party Applications/Devices

   GPS trackers, smart collars, and online resources used separately without integration.

1.3.2.2. Current Processes

1. Health Records

   Pet health information is maintained in paper vaccination booklets.

   These booklets are easily lost or forgotten, and veterinarians rely on them during visits.

2. Care Routines

   Owners rely on memory or mobile phone calendars for reminders of vaccinations, deworming, or grooming.

There is no centralized or reliable reminder system linked to actual health data.

3. Pet Safety and Tracking

   If GPS trackers or smart collars are used, owners must open brand-specific applications.

   Location data is not integrated with other health or care information.

4. Nutrition Decisions

   Owners often make decisions about pet food based on advertisements or general online sources.

   There is no personalized nutritional analysis tailored to the pet's breed, age, or health conditions.

5. Wellness Tracking

   Owners may log weight or feeding data manually in notes, but the information is scattered and unstructured.

   No holistic dashboard exists to present overall pet wellness.

6. Breeding and Sale Proof

   Health certificates are issued on paper by veterinarians.

   Certificates are difficult to validate, can be misplaced, and are vulnerable to forgery or misuse.

1.3.2.3. Weaknesses

1. **Fragmented** Data is scattered across paper, notes, and multiple apps.
2. **Reactive** Owners usually react after problems occur, rather than practicing preventive care.
3. **Lack of Standards** No unified platform integrates health data, reminders, IoT tracking, and nutrition.
4. **Privacy and Security Risks** Paper documents and ad-hoc sharing methods lack proper protection.
5. **High Effort** Owners must manually manage multiple tools and processes, which is inefficient and error prone.

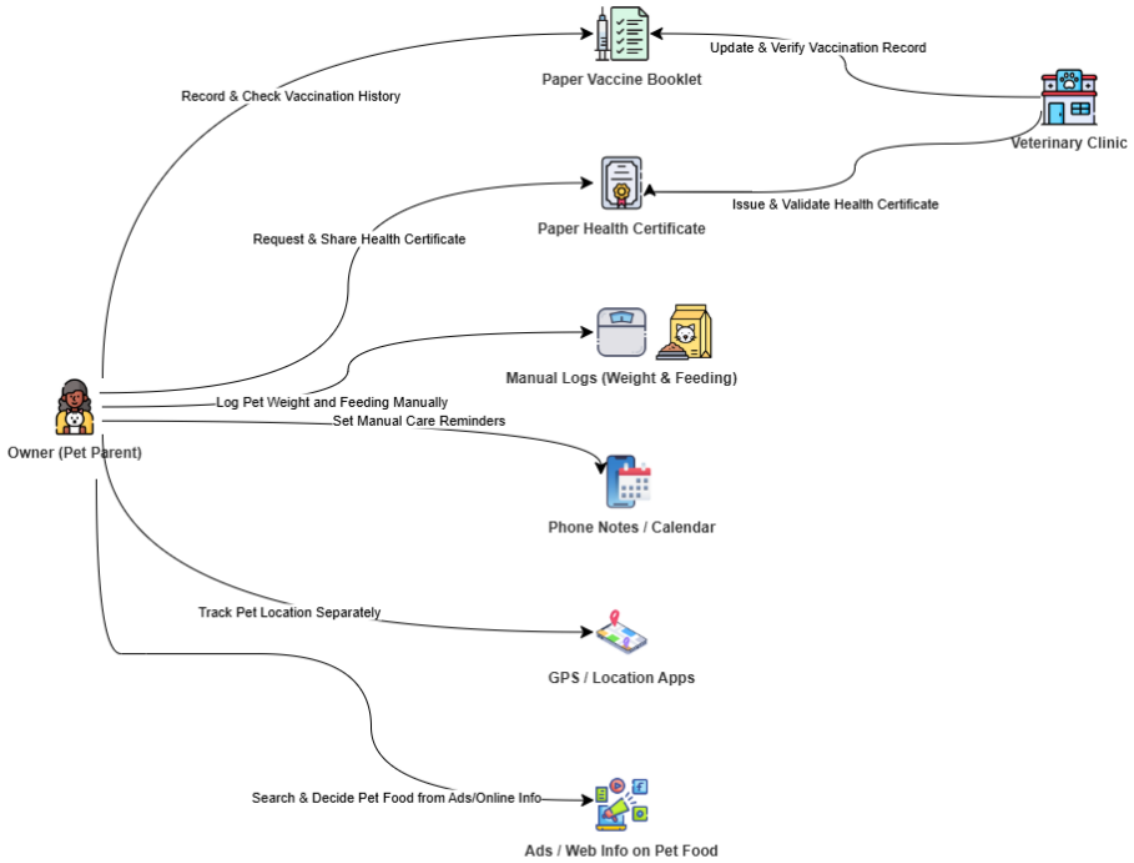


Figure 3 An overview of the as-is situation

1.4 Critical Assumptions and Constraints

1.4.1. Critical Assumptions

Table 1 Project Critical Assumptions

No	Project Critical Assumptions	Description
1	User Engagement	The project assumes that Pet owners will create pet profiles and keep core data up to date (weight, vaccines, diet/activity, feeding logs). AI guidance quality depends on fresh data.
2	Operation and Maintenance	Adequate post-launch resources for cloud capacity, monitoring, security patching, device-compatibility updates, and periodic re-training of the AI nutrition model.
3	Technology Integration	Interoperability with third-party ecosystems (GPS/smart collars, activity trackers, nutrition databases) is feasible via APIs/agreements.
4	Infrastructure Readiness	The platform and AI assistant meet 24/7 availability goals and prompt response times,

		supported by monitoring, autoscaling, and on-call processes.

1.4.2. Constraints

Table 2 Project Constraints

No	Constraints	Description
1	Regulation Compliance	The app Must comply with PDPA/GDPR for owner personal data and pet health data.
2	Development Resources	Limited specialist availability can slow feature delivery; prioritization is required.
3	External Device Integration	Some device vendors restrict APIs or require licensing; direct integration may be limited.
4	Time and Budget	Delivery is capped by funding and a fixed timeline; scope must be managed.

1.5 To-Be System and Function

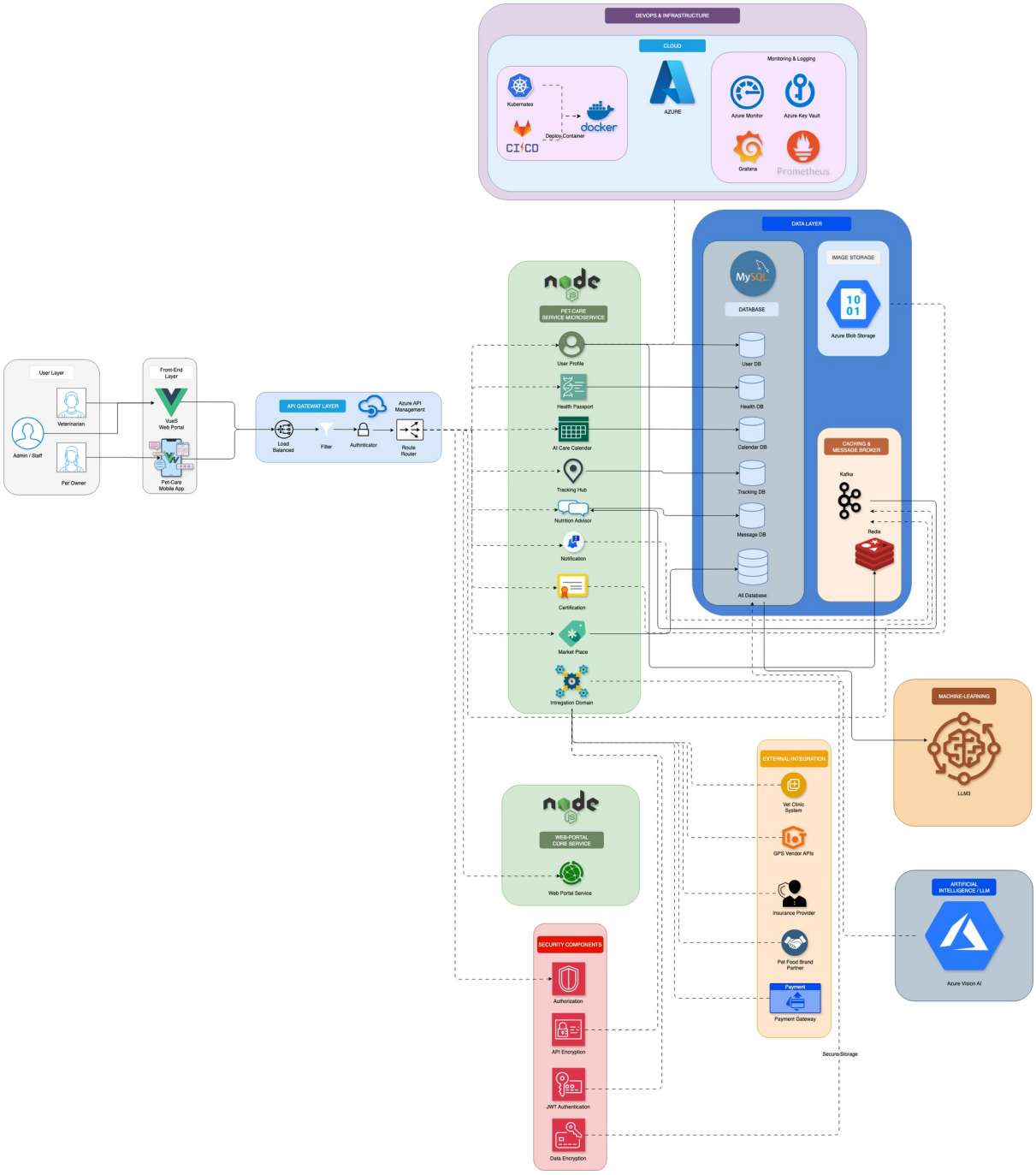


Figure 4 To-be system of PetCare Mobile Application

The To-Be System of the PetCare Mobile Application presents a modernized and unified platform designed to simplify and enrich the experience of pet owners. This upgraded system brings together all essential features into one convenient application. By integrating these capabilities, PetCare aims to make daily pet care easier, more organized, and more enjoyable for every user.

Beyond convenience, the new system fosters trust and collaboration between pet owners, veterinarians, and service providers. The To-Be System transforms PetCare into more than just an app — it becomes a comprehensive lifestyle assistant for pets and their owners, enhancing well-being, safety, and connection across the entire pet ecosystem.

Table 3 PetCare Features

Feature	Feature Name	Brief Description	Phase
F1	Account Registration	Create an account via email/phone/SSO and manage multiple pets under one profile.	1
F2	OTP Verification	One-time passcode (SMS/email or authenticator app) to secure owner & pet data; configurable for login and sensitive actions.	1
F3	Log in	Authenticated sessions with auto timeout and re-authentication when required.	1
F4	Notifications	Push/in-app/email alerts for appointments, medication, and safety events.	1
F5	AI Assistant	Conversational assistant using RAG over each pet's records and curated guidance to answer care questions and propose actions. Users can interact via chat or voice call with the AI for personalized recommendations; non-diagnostic with safety guardrails and escalation to vet when needed.	1
F6	Digital Health Passport	Pet owners can create a secure digital profile for each pet, scan/upload vaccination booklets, and capture QR/label data (brand, batch, expiry). Records include vaccinations, allergies, treatments, medications, and attachments (photos, PDFs). Owners can share a read-only "Digital Health Footprint" with clinics via consented links	1

F7	Intelligent Reminders	An intelligent scheduler generates personalized timelines for vaccinations, deworming, flea/tick prevention, grooming, and medications. Users receive multi-channel reminders (push, in-app, optional SMS/email), can customise frequency/time, and sync events to personal calendars (Google/Apple/Outlook).	1
F8	Safety & Tracking Hub	Integration with popular GPS trackers/smart collars provides live location, geofencing, and escape alerts. Battery health notifications and hybrid tracking (e.g., last-known location, Wi-Fi/Bluetooth assists) increase reliability. Owners see a safety widget and can review safety history	1
F9	AI-Power Nutrition	Scan barcodes/labels to get plain-language nutrition cards and recommendations tailored to age, breed, weight, and conditions. Portion guidance adapts to activity/weight trends.	1
F10	AI-Power Wellness Dashboard	A single view consolidates weight, activity, diet, vaccinations, and meds. Trend analytics highlight anomalies and compute a preventive Wellness Score (e.g., Green/Yellow/Red). A data-completeness checker prompts users to fill gaps (e.g., missing weight logs).	1
F11	Digital Health Certificate	Tamper-resistant digital certificates (e.g., vaccination/health status) are issued by clinics and verifiable via QR. Certificates link to the pet's health passport, with a public read-only verify page to support breeders/sellers and buyer confidence.	1
F12	Vet Portal & Clinic integration	Clinics use role-based templates for intake, treatments, and certificates. OCR tools ease data transition, and the portal supports consented sharing, audit logs, and exports. Vets can log dietary advice, and clinics can	2

		publish post-visit schedules directly to the app.	
F13	Community Marketplace - Loyalty Program	Unified reward system combining partner points, tiers, and perks within the app to boost engagement and retention.	2
F14	Community Marketplace - Adoption	Adoption listings and verified breeders or sellers linked to digital health certificates for trusted transactions.	2
F15	Community Marketplace - Insurance Services	In-app marketplace to compare quotes and buy pet insurance, store e-policies, and receive renewal reminders.	2
F16	Multi-language Support	Provides multilingual UI/UX with locale-aware formats for dates, currencies, and notifications to support global users.	3
F17	Third-Party Integrations & APIs	Adapters connect to food/nutrition databases and multiple wearable/device brands. Future-phase APIs enable insurers, telemedicine, and partner ecosystems to consume verified, consented data	3

Table 4 Problem and Solution Analysis

Problem	Solution	Positive Impact		Negative Impact	
		Organization	Society	Organization	Society
(#P1) Pet health info is scattered & lost	Health Record Digital	Improves credibility via reliable, centralized records; streamlines communication with veterinarians; creates opportunities for premium health-related services.	Owners can better track & manage health, reducing illness; encourages responsible pet care; faster and more accurate veterinary care.	Data storage costs; need security compliance.	Privacy risk if sensitive health data leaks.
(#P2) Owners forget pet care activities (vaccines, deworming, grooming)	Owner-driven Reminder	Enhances app value & credibility; reduces complaints and improves satisfaction; enables premium features (recommended services/products).	Timely care reduces health risks; encourages responsible ownership; lowers incidence of neglected/stray pets.	Higher development/maintenance cost for notification systems.	Over-reliance on app—owners may ignore responsibilities if the app fails.

(#P3) Pets get lost or roam unsafely	PetFit Tracker Integration	Differentiates app with IoT integration; creates partnership opportunities with hardware providers.	Helps locate lost pets quickly; improves overall safety.	Dependence on third-party tracker hardware.	GPS tracking may raise privacy/ethical concerns.
(#P4) Owners unsure about food quality	PetFoodScan	Adds nutrition-helper value; partnership opportunities with pet food brands.	Helps owners select nutritious food; encourages healthy diets.	Ongoing effort to update food database & barcode coverage.	Could discourage local/small brands not present in the database.
(#P5) Hard to see overall pet wellness	Health Dashboard & Analytics	Better UX via clear visualization; opportunity to offer data-driven insights or premium analytics.	Trends easier to understand; enables early detection of health issues.	Higher development/testing costs.	Misinterpretation of graphs may cause stress.
(#P6) Need trusted health proof for breeding/sale	Digital Certificate for Breeding	Builds trust & credibility for breeding/sales; supports regulated, standardized breeding.	Healthier pets and responsible breeding; verifiable proof for buyers.	Careful validation needed (admin overhead).	Risk of misuse if certificates are faked/shared illegally.

1.6 Analysis of options and recommendations

Table 5 Analysis of options

Options	Description
Inhouse Development	**Advantages:** Tight control of roadmap and IP; faster iterations; cultural alignment. **Disadvantages:** Requires hiring across mobile, backend, AI, DevOps, security; higher fixed cost.
Outsource Development	**Advantages:** Rapid capacity; access to diverse expertise; predictable per-sprint cost. **Disadvantages:** Less product ownership; knowledge leakage; vendor lock-in risk.
Hybrid Development	**Advantages:** Core leadership in-house (PO/PM/Tech Lead/UX), delivery pods from vendor(s); best balance of speed, cost, and IP. **Disadvantages:** Requires strong governance (RACI, code ownership, CI/CD, security reviews).

Recommendation: In-House Development with select SaaS add-ons

Rationale: Core development and operations remain in-house to maintain control over quality, CI/CD, and compliance. Security testing is outsourced to certified providers for periodic independent validation, while the OTP service is procured as SaaS to ensure secure and scalable user authentication.

1.7 Preliminary Project Requirements

1.7.1. Justification for Project Phasing

Table 6 Goal in each phase

Phase	Goal	Date
Phase 1: Core Foundation	Attract and onboard pet owners by addressing the most urgent pain points — fragmented health records, forgotten care routines, and lack of trusted sources.	Sep 2025 - 28 Feb 2026
Phase 2: Partnership & Ecosystem Expansion	Strengthen the platform's credibility and stickiness by partnering with key stakeholders across the pet industry.	1 Mar 2026 - 31 Jul 2026
Phase 3: Global Expansion & Advanced Intelligence	Transform PetCare from a regional solution into a global platform with advanced intelligence and an integrated community	1 Aug 2026 - 13 Nov 2026

1.7.2. Functional and Non-Functional Requirement in each phase

Evaluation criteria

Each requirement is scored on **Priority, Difficulty, Risk, Source, Dependencies, and Rationale**.

- Priority: Low, Medium, High Where High is the most important.
- Difficulty: Low, Medium, High Where Low indicates easy implementation.
- Source: Link to Types of User.
- Dependency: Dependency between Feature.
- Rational: Link to problem statement.

Table 7 Software Requirement of The Digital Pet Health Passport (#F1)

ID	Requirement	Type	Priority	Difficulty	Source	Rationale	Dependency
SWR1-01	The system shall support scanning/uploading vaccination	Functional	High	Medium	All	Prevents data loss from paper records and ensures accuracy.	-

	booklets and converting them into digital records.						
SWR1-02	The system shall allow veterinarians to log treatments via a standardized Vet Portal form.	Functional	High	Medium	Vet	Provides consistent and structured records for better use.	F1-01
SWR1-03	The system shall support scanning QR codes/stickers on vaccines to capture brand, batch number, and expiry date.	Functional	High	Medium	Vet	Reduces manual input errors and enables precise vaccine tracking.	F1-02
SWR1-04	The system shall include an AI Assistant that prompts owners (e.g., reminders about medication intake) and updates records automatically.	Functional	Medium	High	All	Improves compliance and continuity of pet care.	F1-02
SWR1-05	The system shall allow pet owners to manually log general health data (e.g., weight, diet, medication).	Functional	Medium	Low	Owner	Provides flexibility for owners to maintain additional details.	F1-01
SWR1-06	The system shall generate a printable and	Functional	High	Medium	All	Facilitates faster vet consultation	F1-03, F1-05

	shareable Digital Health Footprint report.					s and reliable data exchange.	
SWR1-07	The system shall provide a Paper-to-Digital Transition Toolkit for clinics to gradually digitize old records.	Functional	Medium	High	Vet	Eases the transition for clinics still using paper records.	F1-02
SWR1-08	The system shall securely store all health data in a searchable database.	Security	High	High	All	Ensures data privacy and regulatory compliance.	F1-01, F1-02
SWR1-09	The system shall implement role-based access control (Owner vs Vet).	Security	High	High	All	Prevents unauthorized access or modification of sensitive data.	F1-08
SWR-10	The system shall notify pet owners whenever a vet updates the pet's digital records.	Functional	Medium	Low	All	Keeps owners aware of changes in real-time.	F1-02, F1-09

Table 8 Software Requirement of The Intelligent Care Calendar & Reminder System (#F2)

ID	Requirement	Type	Priority	Difficulty	Source	Rationale	Dependency
SWR2-01	The system shall automatically generate a personalized care	Functional	High	Medium	All	Prevents data loss from paper records and	-

	timeline vaccinations, deworming, flea/tick treatment, grooming) using pet data.					ensures accuracy.	
SWR2-02	The system shall send smart notifications in advance (e.g., 1 week before) and on the due date of each care activity.	Functional	High	Medium	Vet	Provides consistent and structured records for better use.	F1-01
SWR2-03	The system shall provide one-time and recurring scheduling options for reminders (e.g., monthly deworming).	Functional	High	Medium	Vet	Increases flexibility to handle unique or repeated pet care tasks.	F2-01
SWR2-04	The system shall allow integration with vet clinics for automated booking when reminders are triggered.	Functional	Medium	High	Vet	Reduces friction by linking reminders with appointment scheduling.	F2-02
SWR2-05	The system shall allow manual customization of reminders by the owner (time,	Functional	Medium	Low	Owner	Empowers owners with control over their pet's care schedule.	F2-01

	frequency, type of activity).						
SWR2-06	The system shall display a Care Calendar view that consolidates upcoming and past activities.	Functional	High	Medium	All	Improves usability by visualizing pet care activities at a glance.	F2-01, F2-02
SWR2-07	The system shall escalate reminders through multiple channels (push notification, email, SMS) for critical care tasks.	Functional	Medium	High	All	Ensures critical reminders are noticed, reducing missed treatments.	F2-02
SWR2-08	The system shall allow syncing with owner's personal calendars (Google, Apple, Outlook).	Security	High	High	Owner	Enhances convenience by integrating with tools owners already use.	F2-06
SWR2-09	The system shall securely store all reminder and scheduling data in the database with encryption.	Security	High	High	All	Protects sensitive health and schedule data from unauthorized access.	F2-01
SWR2-10	The system shall provide reporting/summary of completed vs missed care tasks in the calendar.	Functional	Medium	Low	All	Helps owners and vets monitor compliance and adjust care plans.	F2-06, F2-09

Table 9 Software Requirement of The Centralized Safety & Tracking Hub (#F3)

ID	Requirement	Type	Priority	Difficulty	Source	Rationale	Dependency
SWR3-01	The system shall provide a centralized dashboard with real-time pet location and alerts.	Functional	High	Medium	Owner, Vet	Ensure safety data visible in one place.	-
SWR3-02	The system shall integrate with IoT GPS collars and trackers to update location regularly.	Functional	High	High	Device Vendor	Enable continuous location monitoring.	F3-01
SWR3-03	The system shall generate geofencing alerts when a pet exits a defined safe zone.	Functional	High	Medium	Owner	Prevent pet escape risks.	F3-02
SWR3-04	The system shall log historical movement and safety events in the database.	Functional	Medium	Medium	Owner, Vet	Maintain long-term safety insights.	F3-01, F3-02
SWR3-05	The system shall send critical alerts via multiple channels (push, SMS, email).	Functional	High	Medium	All	Guarantee alerts reach owners.	F3-01, F3-03
SWR3-06	The system shall display a Safety History Report	Functional	Medium	Low	Vet, Owner	Provide safety history for vets.	F3-04

	(escapes, response times).						
SWR3-07	The system shall encrypt all GPS and safety data in transit and at rest.	Security	High	Medium	Security Team	Protect sensitive location data.	-
SWR3-08	The system shall maintain 99.5% uptime for real-time tracking services.	Reliability	High	High	Ops Team	Guarantee service availability.	-

Table 10 Software Requirement of the AI-Powered Nutrition Advisor (#F4)

ID	Requirement	Type	Priority	Difficulty	Source	Rationale	Dependency
SWR4-01	The system shall provide easy-to-understand explanations of nutrition labels.	Functional	High	Medium	Owner	Simplify nutrition understanding.	-
SWR4-02	The system shall link to pet profiles and generate personalized diet plans.	Functional	High	Medium	Owner, Vet	Generate accurate diet recommendations.	F1-01
SWR4-03	The Vet Portal shall allow vets to log dietary recommendations and follow-ups.	Functional	High	Medium	Vet	Ensure structured vet dietary input.	F1-01
SWR4-04	The system shall log meals manually, via	Functional	High	Medium	Owner	Track compliance with feeding plans.	F1-01

	chatbot, or IoT-enabled feeders.						
SWR4-05	The system shall scan food barcodes and provide simplified nutrition cards.	Functional	High	Medium	Owner	Simplify product evaluation.	F4-01
SWR4-06	The system shall generate preventive nutrition alerts from trends in weight/activity.	Functional	High	Medium	Owner, Vet	Enable preventive dietary care.	F4-04
SWR4-07	The system shall rank pet foods based on verified nutritional data and standards.	Functional	Medium	Hard	Admin Vet	Reduce reliance on marketing claims.	F4-05

Table 11 Software Requirement of AI-Powered Wellness Dashboard (#F5)

ID	Requirement	Type	Priority	Difficulty	Source	Rationale	Dependency
SWR5-01	The system shall provide a dashboard with activity, diet, weight, vaccines, and meds.	Functional	High	Low	Owner	Single view of pet wellness.	-
SWR5-02	The system shall compute a Wellness Score with thresholds and trend alerts.	Functional	High	Medium	Owner, Vet	Enable preventive health monitoring.	F1-01

SWR5-03	The system shall visualize long-term health trends and highlight anomalies.	Functional	Medium	Medium	Owner	Help owners understand pet status.	F5-01
SWR5-04	The system shall monitor data completeness and prompt owners to fill missing records.	Functional	Medium	Medium	Owner	Encourage complete data logging.	F1-01
SWR5-05	The system shall detect patterns (e.g., weight gain) and generate predictive alerts.	Functional	Medium	Medium	Owner, Vet	Provide early warning of risks.	F5-04
SWR5-06	The system shall calculate and display a preventive wellness score (Green/Yellow/Red).	Functional	High	Medium	Owner, Vet	Summarize overall wellness.	F5-05

Table 12 Software Requirement of Digital Health Certificate (#F6)

ID	Requirement	Type	Priority	Difficulty	Source	Rationale	Dependency
SWR6-01	The system shall store breeder-submitted health data digital signatures.	Security	High	Medium	Breeder	Prevent falsification of breeder records.	-
SWR6-02	The system shall issue blockchain-	Security	High	High	Breeder	Guarantee immutable trusted certificates.	F6-01

	backed digital certificates.						
SWR6-03	The system shall provide automated workflows for certificate renewal.	Functional	Medium	Medium	Vet, Breeder	Ensure validity over time.	F6-01
SWR6-04	The system shall link certificates directly with the pet's health passport.	Functional	High	Medium	Vet	Ensure certificates reflect health history.	F1-01
SWR6-05	The system shall assign verified breeder badges with QR-based certificate validation.	Functional	Medium	Medium	Buyer	Build buyer trust and verification.	F6-02
SWR6-06	The system shall digitize paper certificates into secure digital formats.	Functional	Medium	Medium	Vet, Breeder	Support transition from paper to digital.	F6-01

1.8 Budget Estimation and Financial Analysis

1.8.1. Profit & Loss Statement (Cost analysis & Benefit analysis & Cashflow analysis)

1.8.1.1. Cost Analysis: CAPEX

Since the PetCare development team operates in a fully remote (work-from-home) environment, there is no requirement for physical office rental, utilities, or on-site infrastructure. Therefore, the capital expenditure (CAPEX) focuses primarily on essential hardware and testing devices needed to support software development, mobile testing, and system deployment. All collaboration, communication, and project management activities are conducted through cloud-based tools and shared platforms.

Table 13 PetCare CAPEX

CAPEX	Unit price	Units (Y0/Y1/Y2)	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Laptop	35,000	15/0/0	525,000	-	-	-	-	-	525,000
Mobile Devices	30,000	4/0/0	120,000	-	-	-	-	-	120,000
Total CAPEX	-	20/0/0	645,000	-	-	-	-	-	645,000

(Unit: THB)

1.8.1.2. Cost Analysis: OPEX

Table 14 PetCare OPEX Summary

Operation Expenses (OPEX)	Head Counts/ Units	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
CTO/Product Owner	1	480,000	1,440,000	1,483,200	1,527,696	1,573,527	1,620,733	8,125,156
Management Team	1	280,000	840,000	865,200	890,156	916,000	943,000	4,734,356
Development and Innovation Team	11	1,980,000	5,940,000	6,188,200	6,299,746	6,487,018	6,680,668	33,575,632
Marketing Team	1	120,000	360,000	370,800	381,924	393,382	405,183	2,031,289
Sales Team	1	120,000	360,000	370,800	381,924	393,382	405,183	2,031,289
Operation Team	1	0	300,000	309,000	318,270	327,818	337,653	1,592,741
Consultant	1	0	21000	9000	0	0	0	30,000
Software Tools	-	106,019	318,061	324,421	330,911	337,529	344,281	1,761,222
Total	-	3,086,019	9,579,061	9,920,621	10,130,627	10,428,656	10,736,701	53,881,685

(Unit: THB)

Table 15 PetCare OPEX Breakdown

Operation Expenses (OPEX)	Head Counts /Units	Monthly Remuneration (THB)	Year 0 (Sep – Dec)	Year 1	Year 2	Year 3	Year 4	Year 5	Total
CTO/Product Owner	1	120,000	480,000	1,440,000	1,483,200	1,527,696	1,573,527	1,620,733	8,245,156
Management Team (3% Remuneration Increase per year)									
Project Manager	1	70,000	280,000	840,000	865,200	890,156	916,000	943,000	4,804,356
Development and Innovation Team (3% Remuneration Increase per year)									
Technical Lead	1	70,000	280,000	840,000	865,200	890,156	916,000	943,000	4,804,356
Senior Business Analyst	1	50,000	200,000	600,000	618,000	636,540	655,636	675,305	3,435,481
Senior System Analyst	1	50,000	200,000	600,000	618,000	636,540	655,636	675,305	3,435,481
Senior UX/UI Designer	1	40,000	160,000	480,000	494,400	509,232	524,509	540,244	2,748,385

Senior Fullstack Developer	1	55,000	220,000	660,000	679,800	700,194	721,200	742,836	3,779,030
Junior Fullstack Developer	2	60,000	240,000	720,000	741,600	763,848	786,764	810,367	4,122,579
Senior Software Quality Assurance Engineer	1	40,000	160,000	480,000	494,400	509,232	524,509	540,244	2,748,385
Senior DevOps Engineer	1	60,000	240,000	720,000	741,600	763,848	786,764	810,367	4,122,579
Security Specialist (Outsource)	1	–	–	–	70,000	–	–	–	70,000
Data Scientist & AI Engineer	1	40,000	160,000	480,000	494,400	509,232	524,509	540,244	4,804,356

Marketing Team (3% Remuneration Increase per year)									
Marketing Officer	1	30,000	120,000	360,000	370,800	381,924	393,382	405,183	2,061,289
Sales Team (3% Remuneration Increase per year)									
Sales Officer	1	30,000	120,000	360,000	370,800	381,924	393,382	405,183	2,061,289
Operation Team (3% Remuneration Increase per year)									
Customer Support	1	25,000		300,000	309,000	318,270	327,818	337,653	1,617,741
Consultant									
Veterinary Advisor	1	–	–	21,000	9,000	–	–	–	–
Software Tools & Package (Monthly Subscriptions)									
Microsoft 365	15	1,485	5,940	17,820	18,176	18,540	18,911	19,290	100,162
GitHub Team/Pro	12	8,316	33,264	99,792	101,788	103,823	105,899	108,017	560,899
Docker Subscription	6	2,970	11,880	35,640	36,353	37,080	37,822	38,578	200,323
Azure AI Vision	1	4,950	19,801	59,404	60,592	61,804	63,040	64,301	333,892

Azure Kubernetes Service	3	713	2,851	8,556	8,727	8,902	9,080	9,262	48,091
Azure Monitoring	50 GB plan	1,073	4,290	12,870	13,127	13,390	13,658	13,931	72,339
Azure API Management	500 GB	330	1,320	3,960	4,039	4,120	4,203	4,287	22,259
Azure Key Vault – Secret Ops	1	10	40	120	122	125	127	130	674
Azure Blob Storage	1	6,600	26,400	79,200	80,784	82,400	84,048	85,729	445,161
Domain Name Registration	1	58	233	699	713	727	741	756	3,927
Total OPEX	-	796,505	3,086,019	9,579,061	9,920,621	10,130,627	10,428,656	10,736,701	53,881,685

(Unit: THB)

Table 16 PetCare Total Cost Analysis

Cost	Details/Description	Total (THB)
CAPEX	Hardware (laptops, mobile test devices)	645,000
OPEX	Management, Dev & Innovation, Marketing, Operation) , Consult, Tools	53,881,685
Reserves (15% of Total Estimate)	Emergency and hidden expenses reserve	8,178,993
Total CAPEX + OPEX	CAPEX + OPEX	54,526,685
Total	CAPEX + OPEX + Reserves	62,705,678

1.8.1.3. Benefit Analysis

PetCare's revenues come from five streams:

1. Pet Parent Subscription plan: premium access with unlimited AI-Power function at 199 THB per month.

Table 17 Pet Parent Subscription plan

Feature	Free Plan	Premium Plan
AI Assistance Chat	Limited Chat (5 Request/Day)	Unlimited Chat
AI Assistance Call	Only Chat with AI Assistance	Call to AI Assistance
AI Reminder	No	Yes
Generate Digital Health Passport	99 THB/Time	Unlimited

2. Breeder Subscription plan: 299 THB per month.

Table 18 Breeder Subscription plan

Feature	Free Plan	Subscription Plan
Paper-to-Digital Certificate Converter	No	Yes

3. Clinic subscription plan for using Web Portal: 1,500 THB per clinic per month.

1.8.1.4. Cost-Benefit Analysis

Table 19 PetCare Cost-Benefit Analysis

PetCare Cost-Benefit Analysis							
Discount Rate	10%						
Project Timeline (Year)	**0**	**1**	**2**	**3**	**4**	**5**	**Total**
Discount Factor	1	0.909090909	0.826446281	0.751314801	0.683013455	0.620921323	
1. Costs	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Total**
1.1. CAPEX							
Total CAPEX Costs	645,000	-	-	-	-	-	645,000
Total Discounted CAPEX Costs	645,000	-	-	-	-	-	645,000
Cumulative Discounted CAPEX Costs	645,000	645,000	645,000	645,000	645,000	645,000	3,870,000
1.2. OPEX							
Total OPEX Costs	3,086,019	9,579,061	9,920,621	10,130,627	10,428,656	10,736,701	53,881,685
Total Discounted OPEX Costs	3,086,019	8,708,237	8,198,860	7,611,290	7,122,912	6,666,647	41,393,966
Cumulative Discounted OPEX Costs	3,086,019	11,794,256	19,993,117	27,604,407	34,727,319	41,393,966	95,275,651

1.3. Reserves							
Total Reserves Costs	559,653	1,436,859	1,488,093	1,519,594	1,564,298	1,610,505	11,544,509
Total Costs	4,290,672	11,015,920	11,408,714	11,650,221	11,992,954	12,347,206	88,507,906
Total Discounted Costs	4,290,672	10,014,473	9,428,689	8,752,983	8,191,349	7,666,643	62,237,790
Cumulative Discounted Costs	4,290,672	14,305,145	23,733,834	32,486,817	40,678,166	48,344,810	281,598,013
2. Benefits	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
2.1 Standard	-	-	3,568,044	9,695,368	19,527,172	32,577,734	65,368,318
2.2 Premium	-	-	1,195,355	3,340,915	6,944,512	11,787,216	23,267,998
2.3 Breeder Users	-	-	35,921	104,578	204,148	344,370	689,017
2.4 Clinic	-	-	600,681	1,259,139	2,124,168	3,290,680	7,274,668
Total Benefits	-	-	5,400,001	14,400,000	28,800,000	48,000,000	96,600,001
Total Discounted Benefits Costs	-	-	4,462,811	10,818,933	19,670,788	29,804,224	64,756,755
Cumulative Discounted Benefits Costs	-	-	4,462,811	15,281,744	34,952,531	64,756,755	119,453,841
3. Financial Analysis							
3.1. NPV	2,518,965						
3.2. ROI	4.05%						
3.3. Payback years	4.2 ≈ 5						

3.4. IRR	11%
3.5. Life Cycle Cost	62,237,790

(Unit: THB)

1.8.2. Financial Analysis

This analysis covers the 14.5-month window (Years 0–1) and applies a 10% discount rate. Year 2 is out of scope in this document. The analysis includes capital expenditures (CAPEX) and operating expenses (OPEX)—including software tools and software packages—and uses the updated benefits breakdown for consumer and business segments. Advertising and partner revenue are assumed to be zero in Years 0 through 2. All figures are presented in million Thai Baht (THB)

Results

- Net Present Value (NPV, 10%): ≈ 2,518,965 THB
- Return on Investment (ROI, undiscounted): ≈ 4.05%%
- Payback Period: 4.2 ≈ 5
- Internal Rate of Return (IRR): ≈ 11%

1.9. Schedule Estimation

The project is structured into five phases over a 14.5 months delivery timeline (1 Sep 2025 – 13 Nov 2026): Initiating, Planning, Executing, Monitoring & Controlling, and Closing. The project will follow the SDLC **Agile framework** (Scrum) with **Four-week sprints.** Further details are provided in a later section.

1.10. Development Process

PetCare will be delivered with an Agile SDLC (Scrum) and reported using the PMBOK process groups—Initiation, Planning, Executing, Monitoring & Controlling, and Closing. Work is time-boxed into 4-week sprints, each producing a potentially shippable increment that meets our Definition of Ready/Done, security and quality gates.

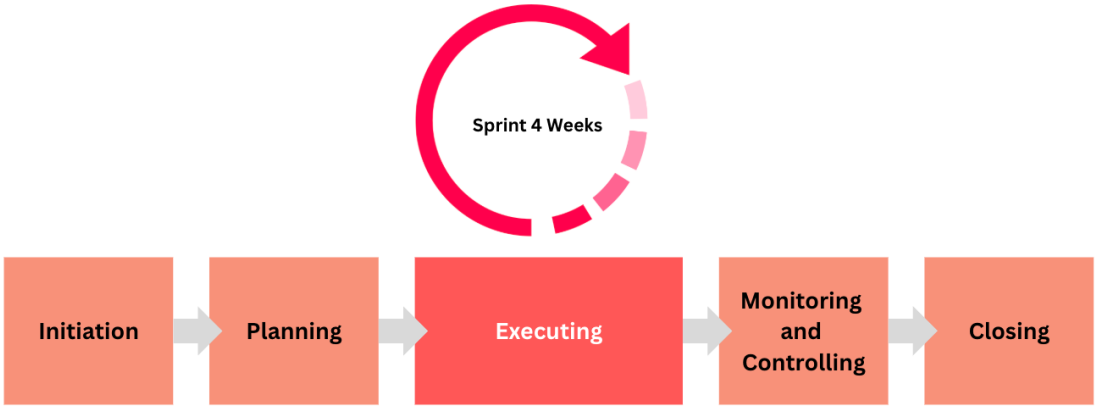


Figure 5 PetCare SDLC

Table 20 PetCare SDLC Breakdown

PetCare SDLC (Agile)			
Step	**Process Name**	**Activities**	**Detail**
1	Initiation	Project Charter	Define project objectives, scope, deliverables, and key success metrics for PetCare.
		Stakeholder Register	Identify and document all stakeholders, their roles, influence, and communication needs.
		Kickoff Meeting	Conduct initial meeting to align team members, confirm goals, and finalize the project roadmap.
2	Planning	Requirements Elicitation	Collect user needs and system requirements from pet parents, vets, and partners.
		Stakeholder Interviews	Conduct structured interviews to validate expectations and gather domain insights.
		Competitor Research	Analyze competing PetTech apps to benchmark features, UX, and pricing.
		Define Release Plan and Scope of Work	Outline sprint timeline, milestones, and Definition of Done (DoD) to align scope and quality.
		UX Wireframes	Design initial user interface layouts for onboarding, health records, and dashboards.
		Architecture Baseline	Establish system architecture (frontend, backend, database, API layers).
		Test & Security Planning	Plan testing strategy, identify key risks, and define security baselines (authentication, consent).

<table>
<tr><td></td><td></td><td>Tooling Setup (CI/CD, Jira, GitHub)</td><td>Configure project management tools, GitHub repository, and CI/CD pipelines.</td></tr>
<tr><td rowspan="13">3</td><td rowspan="13">Executing</td><td>Sprint 1: Onboarding (F1-F5)</td><td rowspan="13">Activities for each sprint
1. Sprint Planning
- Product backlog
- Sprint backlog
- Daily Scrum
2. Sprint Execution (Analysis & Design, Implementation, Evaluation, Deployment)
3. Sprint Review (Demo)
4. Sprint Retrospective
5. Backlog Refinement (Groom next stories, re-estimate, surface risks/dependencies.)</td></tr>
<tr><td>Sprint 2: Digital Health Passport (F6)</td></tr>
<tr><td>Sprint 3: Intelligent Reminders (F7)</td></tr>
<tr><td>Sprint 4: Safety & Tracking Hub (F8)</td></tr>
<tr><td>Sprint 5: AI-Powered Nutrition (F9)</td></tr>
<tr><td>Sprint 6: AI-Powered Wellness Dashboard (F10)</td></tr>
<tr><td>Sprint 7: Digital Health Certificate (F11)</td></tr>
<tr><td>Sprint 8: Vet Portal & Clinic integration (F12)</td></tr>
<tr><td>Sprint 9: Community Marketplace - Loyalty Program (F13)</td></tr>
<tr><td>Sprint 10: Community Marketplace - Adoption (F14)</td></tr>
<tr><td>Sprint 11: Community Marketplace - Insurance Services (F15)</td></tr>
<tr><td>Sprint 12: Multi-language Support (F16)</td></tr>
<tr><td>Sprint 13: Third-Party Integrations & APIs (F17)</td></tr>
<tr><td rowspan="3">4</td><td rowspan="3">Monitoring and Controlling</td><td>Security Monitoring</td><td>Dependency scans, permission reviews, consent & retention audits.</td></tr>
<tr><td>Risk Monitoring</td><td>Maintain RAID log, assess risk likelihood and impact, and define mitigation plans.</td></tr>
<tr><td>Progress Monitoring</td><td>Update burndown chart and velocity metrics to forecast sprint completion.</td></tr>
<tr><td>5</td><td>Closing</td><td>Create Project Documents</td><td>Compile and finalize all project artifacts.</td></tr>
</table>

		Release Candidate Build	Prepare final system version; perform regression and security validation.
		Regression & Security Testing	Validate functional integrity and security before deployment.
		Lessons Learned Workshop	Document team insights and best practices for continuous improvement.
		Project Sign-Off	Sponsor acceptance and formal close.

1.11. Potential Risk

Table 21 Potential Risk Identification

Risk Area	Description	Probability	Impact	Mitigate Strategy
AI guidance accuracy	Nutrition advice inaccurate or unsafe.	Medium	High	Curated rules, evaluation dataset, human-review for edge cases, clear disclaimers
Data quality	Incomplete or messy records from users.	High	Medium	Required fields, OCR capture, field validation, duplicate detection, prompts with examples.
Third-party integration	GPS or food databases have limited APIs or change terms.	Medium	High	Adapter pattern, multiple vendors, contracts, monitoring of API changes, graceful fallback.
Security incident	Unauthorized access or data leak.	Low	High	Secure coding, secrets management, 2FA, logging, penetration test, incident plan.

2. Project Charter

Table 22 PetCare Project Charter

Project Charter	
Project Title:	PetCare Mobile Application Development
Project Start Date:	1 Sep 2025
Project Finish Date:	13 November 2026
Budget Information:	The total project expenditure for the 14.5-month period amounts to **13,310,080 THB**, consisting of **capital expenditures (CAPEX)** of **645,000 THB** and **operating expenditures (OPEX)** for **Years 0–1** totaling **12,665,080 THB**.
Financial Analysis:	**Financial Analysis:** **-** Net Present Value (NPV, 10%): ≈ 2,518,965 THB - Return on Investment (ROI, undiscounted): ≈ 4.05% - Payback Period: 4.2 ≈ 5 years - Internal Rate of Return (IRR): ≈ 11% - Life Cycle Cost (CAPEX+OPEX, THB): ≈ 62,237,790 THB
Project Manager:	Mr. Pongsakorn Rattanakul (Project Manager)
Project Objectives:	Deliver a consolidated pet health platform that solves fragmented pet health records, missed care appointments, safety/nutrition concerns, and lack of an overall health view; create preventive insights and verifiable digital health certification in Thailand, under Agile delivery.
Success Criteria:	1. Net Promoter Score (NPS) ≥ 50 2. Crash-free Sessions ≥ 99.8% 3. Uptime ≥ 99.5% 4. Performance Test > 95% 5. Retention ≥ 40%
Approaches:	**Requirements Definition** Study the existing information and related systems or applications to identify deficiencies and determine the necessary capabilities for system development. 2. Problem Analysis and Requirements Specification (SRS)

	Conduct meetings with stakeholders to review the documented requirements and confirm that they align with stakeholder expectations. 3. Software Prototype Presentation Organize a meeting to introduce the software prototype to stakeholders and gather their feedback and suggestions for improvement. 4. Software Development using Agile Scrum Framework Develop the system under the Agile Scrum framework to ensure valuable features or deliverables can be released to users efficiently and iteratively. 5. Quality Assurance Perform verification and validation of the developed product to detect defects and evaluate its quality before final delivery. 6. Software Maintenance and Improvement Address any defects or bugs identified after deployment and continuously enhance the system's performance and stability.

3. Stakeholder Register

The PetCare project engages a diverse group of stakeholders from both internal and external domains to ensure a holistic perspective across business, technology, and user needs.

To support effective collaboration and accountability, stakeholders are categorized into four primary groups:

1. **Users** – Individuals who directly interact with the system and contribute feedback to improve its functionality and usability.
2. **Ecosystem Partners** – External organizations that collaborate with the project to expand service integration and strengthen the overall ecosystem.
3. **Development Team** – Internal members responsible for the planning, execution, and continuous improvement of the PetCare platform.
4. **Consultants & Advisors** – External specialists who provide expert guidance and ensure alignment with professional and regulatory standards.

Table 23 Stakeholders Register for PetCare

No	Name	Position	Internal/ External	Project Role	Contact Information
1	Mr. Napat	Pet Owner Representative (End-user)	External	Keep health records, configure reminders,	Napat.Cha@gmail.com

	Chaiyapat			consent to data use, safety tracking, provide feedback	
2	Dr. Kittipong Wattanakul	Veterinarians	External	Record medical history, issue digital health certificates, onboarding/training, data privacy compliance	Kittipong.wat@gmail.com
3	Mr. Pradit Thongchai	Breeder/Seller Representative	External	Use verifiable health certificates, comply with standards and policies	Pradit.tho@gmail.com
4	Ms. Sirinapa Boonmee	Pet-food Brands/ Ecosystem Partners	External	Co-develop nutrition features/campaigns, receive anonymized insights under consent	Sirinapa.boo@pfoodth.co.th
5	Mr. Thanawat Srisa-ard	Insurer Partner (Phase 2)	External	Explore products using verified health data, ensure regulatory compliance	Thanawat.sri@petcare-thailand.com
6	Mr. Anuwat Phasuk	Chief Technology Officer (CTO)/Product Owner (PO)	Internal	Define product vision and roadmap, approve scope and budget, set technical direction, ensure alignment across teams, and oversee governance and delivery quality.	Anuwat.pha@petcare-thailand.com
7	Mr. Pongsakorn Rattanakul	Project Manager (PM)	Internal	Manage scope/schedule/budget/risks, facilitate Scrum events, stakeholder comms	Pongsakorn.rat@petcare-thailand.com

8	Mr. Teerapat Kittisak	Technical Lead	Internal	End-to-End Architecture & quality gates, reviews, security baseline, integration strategy	Teerapat.kit@petcare-thailand.com
9	Mr. Kittinan Suwankiri	Senior Fullstack Developer	Internal	Build APIs/data models/integrations, unit/integration tests, performance	Kittinan.suw@petcare-thailand.com
10	Ms. Ratchada Pholsri	Junior Fullstack Developer	Internal	Build APIs/data models/integrations, unit/integration tests, performance	Ratchada.pho@petcare-thailand.com
11	Mr. Wichan Dechasuk	Junior Fullstack Developer	Internal	Build APIs/data models/integrations, unit/integration tests, performance	Wichan.dec@petcare-thailand.com
12	Ms. Warunee Phrompanit	Senior Business Analyst	Internal	Elicits/clarifies requirements; writes user stories & acceptance criteria; process mapping; aligns PO, users, and engineering.	Warunee.phr@petcare-thailand.com
13	Mr. Chanin Charoensuk	Senior System Analyst	Internal	Detailed solution specs: APIs, sequence/ER diagrams, interface contracts; non-functional specification hand-off to dev/QA.	Chanin.cha@petcare-thailand.com
14	Mr. Arthit	Data Scientists/AI Engineer	Internal	Rules/models for nutrition, wellness dashboards,	Arthit.boo@petcare-thailand.com

	Boonyasiri			pipelines, privacy-aware analytics	
15	Ms. Jiraporn Sangsuk	Senior Software QA Engineer	Internal	Test strategy/automation, UAT support, defect management, DoD compliance	Jiraporn.san@petcare-thailand.com
16	Ms. Pimchanok Yodphet	Senior UX/UI Designer	Internal	User research, IA/wireframes, prototyping, usability, Thai-first UX	Pimchanok.yod@petcare-thailand.com
17	Mr. Natthakit Thanasri	DevOps Engineer	Internal	Manage CI/CD pipelines, cloud deployment, monitoring, and system reliability.	Natthakit.tha@petcare-thailand.com
18	Ms. Weeraphon Sutham	Security Specialist	External	PDPA consent flow, encryption/RBAC, reviews & audits	Weeraphon.sut@gmail.com
19	Dr. Supawadee Klongchai	Veterinary Advisor (Consultant)	External	Validate clinical flows and certificate criteria, content guidance	Supawadee.Klo@gmail.com
20	Ms. Thanyaporn Kanchana	Marketing Officer	Internal	Acquisition, clinic bundles, content & campaigns, partner co-marketing	Thanyaporn.kan@petcare-thailand.com

21	Mr. Aekkarat Prempree	Customer Support	Internal	Ticketing/help center, VOC loop to product, education materials	Aekkarat.pre@petcare-thailand.com
22	Ms. Orathai Chantara	Sales Officer	Internal	Incident triage, data ops, compliance admin, routine maintenance	Orathai.cha@petcare-thailand.com
23	Mr. Thanakorn Phasuk	Chief Executive Officer (CEO)	Internal	Approve budget and strategy, ensure project alignment with business goals and compliance.	Thanakorn.pha@petcare-thailand.com

4. Project Organization Chart

The PetCare Project Organization is structured into **seven primary groups** to ensure strategic alignment, operational efficiency, and high-quality product delivery across all business and technical domains:

- **Management Team** – Provides executive strategy, project direction, and governance oversight.
- **Development & Innovation Team** – Serves as the core delivery group, responsible for building and maintaining application features end-to-end.
- **Marketing Team** – Manages go-to-market and customer support.
- **Sales Team** - Oversees client acquisition, partner relationships, and revenue operations.
- **Operations Team** – Ensures application operations and maintenance.
- **Financial Team** – Oversees budgeting, expense tracking, and financial reporting for the project.
- **Human Resource Ream** – Manages resource allocation, hiring, and training coordination.

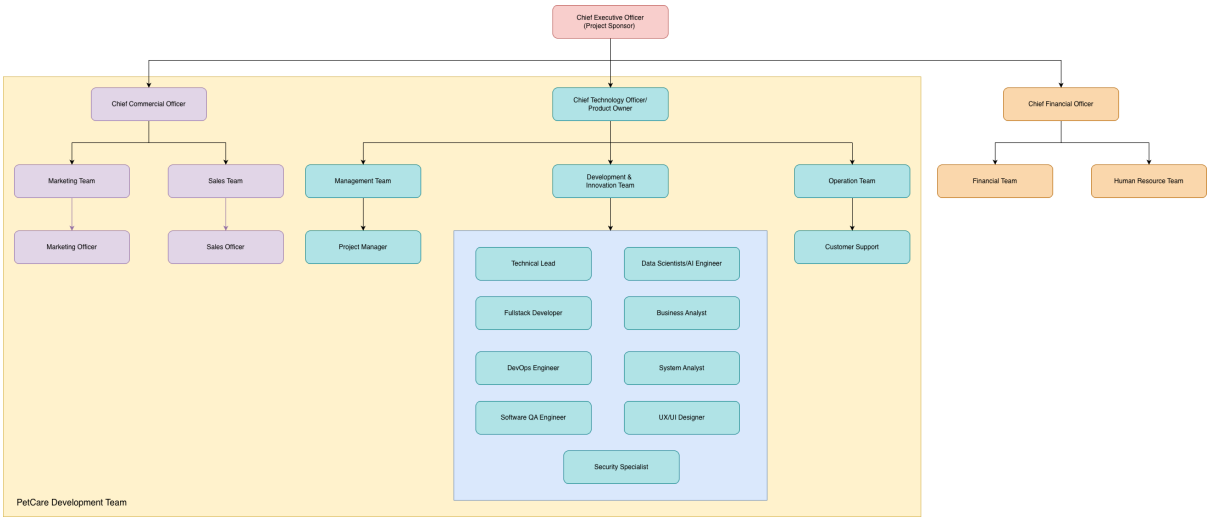


Figure 6 PetCare Team Organization Structure

Table 24 Key responsibilities of each role

Group	Role	Key Responsibilities
-	Chief Executive Officer (CEO)	Provides executive oversight, approves project funding, ensures alignment with business goals, and resolves high-level issues.
-	Chief Technology Officer (CTO)/Product Owner (PO)	Defines product vision and roadmap, sets technical direction, approves scope and priorities, and ensures delivery quality.
-	Chief Commercial Officer (CCO)	Leads commercial strategy, oversees marketing and sales performance, and drives customer and partner engagement.
Management Team	Project Manager (PM)	Oversees project scope, schedule, budget, risks, and communication. Owns Gantt chart, sprint reviews, and reporting.
Development & Innovation Team	Technical Lead	Defines system architecture, ensures code quality, and supports cross-module technical decisions.
	Fullstack Developer	Implements app features, APIs, authentication, offline mode, and database schema integrations.
	Business Analyst	Conducts stakeholder interviews, gathers requirements, defines user stories, acceptance criteria, NFRs, and KPIs.

	System Analyst	Translates business stories into system specifications, workflows, and interface contracts.
	Data Scientist/AI Engineer	Designs nutrition models, manages ML pipelines, evaluates accuracy, and supports AI-driven features
	DevOps Engineer	Builds and maintains CI/CD pipelines, infrastructure, and observability systems.
	Software QA Engineer	Develops test strategies, executes test cases, automates testing, and supports UAT.
	Security Specialist	Performs threat modeling, ensures data protection, manages access controls, and oversees consent flows.
	UX/UI Designer	Conducts user research, creates wireframes, prototypes, and defines design systems.
Marketing Team	Marketing Officer	Leads positioning, launch campaigns, and partner co-marketing initiatives.
Sales Team	Sales Officer	Manages partner pipelines, contract renewals, and supports commercial deals.
Operation Team	Customer Support	Handle production issues, coordinate with Dev team for unresolved cases
Financial Team	-	Oversees budgeting, expense tracking, and financial reporting for the project.
Human Resource Team	-	Manages resource allocation, hiring, and training coordination.

5. Scope Baseline

5.1. Scope statement

Table 25 Scope Statement

Scope statement
Project Title: PetCare Mobile Application Development **Date: Project Timeline: 1 Sep 2025 - 13 Nov 2026**
Project Justification: PetCare Project Management Team
Product Characteristics and Requirements: 1. Mobile app (iOS/Android) for pet owners and veterinary clinics 2. Web application for desktop access and administration

3. AI-powered health and nutrition analysis with automatic routine reminders
4. Digital Health Certificate system compliant with PDPA/GDPR security standards
5. Vet Portal for clinics to issue certificates and update health records
6. Pet Profile management (multi-pet per account) and medical history
7. Health Record management (vaccinations, treatments, attachments)
8. Scheduling and intelligent reminders for visits, medication, and vaccines
9. Safety & Tracking Hub with IoT/GPS device integration and alerts
10. AI-powered Wellness Dashboard with predictive analytics
11. Payment Gateway for subscriptions and in-app services
12. Community Marketplace with reviews, coupons, and Loyalty Program
13. Adoption module for listings and approval workflows
14. Insurance services integration (quote, plan selection, claim via partner APIs)
15. Multi-language support (e.g., Thai/English) with AI translation
16. Third-party integrations via API Gateway and adapter services (extensible)

Product User Acceptance Criteria:

1. Net Promoter Score (NPS) ≥ 50
2. Crash-free Sessions ≥ 99.8%
3. Uptime ≥ 99.5%
4. Performance Test > 95%
5. Retention ≥ 40%

Summary of Project Deliverables

Project management-related deliverables: business case, charter, team contract, scope statement, WBS, schedule, cost baseline, status reports, final project presentation, final project report, lessons-learned report, and any other documents required to manage the project.

Product-related deliverables: research reports, design documents, software code, hardware, etc.

1. Software Requirements Specification document
2. System Design Specification document
3. User acceptance test document
4. Requirement Traceability Matrix
5. Performance Test Document
6. Uptime Monitoring Report

Assumptions:

1. Users consent to share pet health data for AI analysis.

2. IoT and wearable device APIs are available for integration. 3. Cloud and third-party services remain stable during the project. 4. Pet owners regularly update data to maintain accuracy. 5. Partnerships with clinics and ecosystem partners are established by Phase 2.
Constraints: Project duration is fixed between 1 September 2025 and 13 November 2026 (≈14.5 months). All budget, schedule, and analysis components in this document are based on this timeframe.

5.2. WBS

Table 26 PetCare WBS

5.2.1.1. Initiation 1.1. Project Charter 1.2. Stakeholder Register 1.3. Kickoff Meeting 5.2.1.2. Planning 2.1. Requirements Elicitation 2.2. Stakeholder Interviews 2.3. Competitor Research 2.4. Define Release Plan and Scope of Work 2.5. UX Wireframes 2.6. Architecture Baseline 2.7. Test & Security Planning 2.8. Tooling Setup (CI/CD, Jira, GitHub) 5.2.1.3. Executing 6.1. Sprint 1: Onboarding 6.1.1. Sprint Planning 6.1.2. Develop Account Registration Page (UI, Frontend, APIs) 6.1.3. Develop Log in Page (UI, Frontend, APIs) 6.1.4. Develop Notifications (UI, Frontend, APIs) 6.1.5. Develop Onboarding Page (UI, Frontend) 6.1.6. Develop Setting Page (UI, Frontend, APIs) 6.1.7. Develop AI Assistant Page (UI, Frontend, Backend, APIs) 6.1.8. Develop Payment Gateway (APIs)

6.1.9. Develop Landing Page (UI, Frontend, APIs)
6.1.10. Sprint Review
6.1.11. Sprint Retrospective
6.1.12. Backlog Refinement
6.2. Sprint 2: Digital Health Passport
6.2.1. Sprint Planning
6.2.2. Develop Add Pet Profile Page (UI, Frontend, Backend, APIs, Databases)
6.2.3. Develop Manage Health Record Page (UI, Frontend, Backend, APIs, Databases)
6.2.4. Sprint Review
6.2.5. Sprint Retrospective
6.2.6. Backlog Refinement
6.3. Sprint 3: Intelligent Reminders
6.3.1. Sprint Planning
6.3.2. Develop Schedule Page (UI, Frontend, Backend, APIs, Databases)
6.3.3. Develop Manage Reminder Page (UI, Frontend, Backend, APIs, Databases)
6.3.4. Sprint Review
6.3.5. Sprint Retrospective
6.3.6. Backlog Refinement
6.4. Sprint 4: Safety & Tracking Hub
6.4.1. Sprint Planning
6.4.2. Develop Manage Safety & Tracking Hub Page (UI, Frontend, Backend, APIs, Databases)
6.4.3. Sprint Review
6.4.4. Sprint Retrospective
6.4.5. Backlog Refinement
6.5. Sprint 5: AI-Powered Nutrition
6.5.1. Sprint Planning
6.5.2. Develop Manage AI-Powered Nutrition Page (UI, Frontend, Backend, APIs, Databases)
6.5.3. Gathering Nutrition Data (Scraping data and Manage Nutrition DB)
6.5.4. Sprint Review

6.5.5. Sprint Retrospective
6.5.6. Backlog Refinement
6.6. Sprint 6: AI-Powered Wellness Dashboard
6.6.1. Sprint Planning
6.6.2. Develop AI-Powered Wellness Dashboard Page (UI, Frontend, Backend, APIs, Databases)
6.6.3. Develop Wellness Analytics Models
6.6.4. Sprint Review
6.6.5. Sprint Retrospective
6.6.6. Backlog Refinement
6.7. Sprint 7: Digital Health Certificate
6.7.1. Sprint Planning
6.7.2. Develop Manage Digital Health Certificate Page (UI, Frontend, Backend, APIs, Databases)
6.7.3. Sprint Review
6.7.4. Sprint Retrospective
6.7.5. Backlog Refinement
6.8. Sprint 8: Vet Portal & Clinic integration
6.8.1. Sprint Planning
6.8.2. Develop Vet Portal (UI, Frontend, Backend, APIs, Databases)
6.8.3. Sprint Review
6.8.4. Sprint Retrospective
6.8.5. Backlog Refinement
6.9. Sprint 9: Community Marketplace - Loyalty Program
6.9.1. Sprint Planning
6.9.2. Develop Community Marketplace Page (UI, Frontend, APIs)
6.9.3. Develop Loyalty Program Page (UI, Frontend, Backend, APIs, Databases)
6.9.4. Sprint Review
6.9.5. Sprint Retrospective
6.9.6. Backlog Refinement
3.10. Sprint 10: Community Marketplace - Adoption

1.10.1. Sprint Planning

1.10.2. Develop Adoption Page (UI, Frontend, Backend, APIs, Databases)

1.10.3. Sprint Review

1.10.4. Sprint Retrospective

1.10.5. Backlog Refinement

3.11. Sprint 11: Community Marketplace - Insurance Services

3.11.1. Sprint Planning

3.11.2. Develop Insurance Services Page (UI, Frontend, Backend, APIs, Databases)

3.11.3. Sprint Review

3.11.4. Sprint Retrospective

3.11.5. Backlog Refinement

3.12. Sprint 12: Multi language Support

3.12.1. Sprint Planning

3.12.2. Develop AI Translation Integration

3.12.3. Develop Locale-Aware UI

3.12.4. Sprint Review

3.12.5. Sprint Retrospective

3.12.6. Backlog Refinement

3.13. Sprint 13: Third-Party Integrations & APIs

3.13.1. Sprint Planning

3.13.2. Develop API Gateway & Adapters

3.13.3. Sprint Review

3.13.5. Sprint Retrospective

3.13.6. Backlog Refinement

5.2.1.4. Monitoring and Controlling

4.1. Security Monitoring

4.2. Risk Monitoring

4.3. Progress Monitoring

5.2.1.5. Closing

5.1. Create Project Documents

5.2. Release Candidate Build

5.3. Regression & Security Testing

5.4. Lessons Learned Workshop 5.5. Project Sign-Off

5.3. WBS dictionary of two WBS items

Table 27 WBS dictionary of items 1

WBS dictionary
Project Name: PetCare
WBS Item Number: 1.1.
WBS Item Name: Project Charter
Description: Establishes the PetCare project's purpose, objectives, scope, major deliverables, and key stakeholders. Serves as the official authorization to begin the project. **Responsibility:** Project Manager/Project Sponsor **Input:** Business case, stakeholder analysis, feasibility study **Output:** Approved Project Charter, stakeholder alignment, project authorization record **Acceptance Criteria:** Project Charter signed by sponsor; project objectives, scope, and stakeholder roles clearly defined and communicated.

Table 28 WBS dictionary of items 2

WBS dictionary
Project Name: PetCare
WBS Item Number: 4.3.
WBS Item Name: Progress Monitoring
Description: Monitors overall progress of PetCare sprints by tracking schedule adherence, burn-down metrics, and quality performance to ensure timely and efficient delivery. **Responsibility:** Project Manager **Input:** Sprint reports, Jira metrics, risk and issue logs, team status updates **Output:** Updated progress dashboard, burn-down charts, status reports, and performance metrics

Acceptance Criteria:
Progress reports updated weekly; deviations within ±10% of baseline; approved by management during sprint reviews.

6. Schedule Baseline

6.1. Gantt chart

PetCare Gantt Chart Link:

https://drive.google.com/drive/folders/1j4SKlTOjb4emS6WHyq1V0MYtjWQILImG?usp=sharing

	Name	Predecessors	Duration	Start	Finish
1	**PetCare**		**315 days?**	**9/1/25, 8:00AM**	**11/13/26, 5:00PM**
2	**Initiation**		**15 days**	**9/1/25, 8:00AM**	**9/19/25, 5:00PM**
3	Project Charter		5 days	9/1/25, 8:00AM	9/5/25, 5:00PM
4	Stakeholder Register	3	5 days	9/8/25, 8:00AM	9/12/25, 5:00PM
5	Kickoff Meeting	4	5 days	9/15/25, 8:00AM	9/19/25, 5:00PM
6	**Planning**		**25 days**	**9/22/25, 8:00AM**	**10/24/25, 5:00PM**
7	Requirements Elicitation	5	5 days	9/22/25, 8:00AM	9/26/25, 5:00PM
8	Stakeholder Interviews	7	5 days	9/29/25, 8:00AM	10/3/25, 5:00PM
9	Competitor Research	8	5 days	10/6/25, 8:00AM	10/10/25, 5:00PM
10	Release Plan & Define Definition of Done (DoD)	9	5 days	10/13/25, 8:00AM	10/17/25, 5:00PM
11	UX Wireframes	10	5 days	10/20/25, 8:00AM	10/24/25, 5:00PM
12	Architecture Baseline	10	5 days	10/20/25, 8:00AM	10/24/25, 5:00PM
13	Test & Security Planning	10	5 days	10/20/25, 8:00AM	10/24/25, 5:00PM
14	Tooling Setup (CI/CD, Jira, GitHub)	10	5 days	10/20/25, 8:00AM	10/24/25, 5:00PM
15	**Executing**		**260 days**	**10/27/25, 8:00AM**	**10/23/26, 5:00PM**
16	**Sprint 1: Onboarding**		**20 days**	**10/27/25, 8:00AM**	**11/21/25, 5:00PM**
17	Sprint Planning	14	1 day	10/27/25, 8:00AM	10/27/25, 5:00PM
18	Develop Account Registration Page (UI, Frontend, APIs)	17	16 days	10/28/25, 8:00AM	11/18/25, 5:00PM
19	Develop Log in Page (UI, Frontend, APIs)	17	16 days	10/28/25, 8:00AM	11/18/25, 5:00PM
20	Develop Notifications (UI, Frontend, APIs)	17	16 days	10/28/25, 8:00AM	11/18/25, 5:00PM
21	Develop Onboarding Page (UI, Frontend)	17	16 days	10/28/25, 8:00AM	11/18/25, 5:00PM
22	Develop Setting Page (UI, Frontend, APIs)	17	16 days	10/28/25, 8:00AM	11/18/25, 5:00PM
23	Develop AI Assistant Page (UI, Frontend, Backend, APIs)	17	16 days	10/28/25, 8:00AM	11/18/25, 5:00PM
24	Develop Payment Gateway (APIs)	17	16 days	10/28/25, 8:00AM	11/18/25, 5:00PM
25	Develop Landing Page (UI, Frontend, APIs)	17	16 days	10/28/25, 8:00AM	11/18/25, 5:00PM
26	Sprint Review	25	1 day	11/19/25, 8:00AM	11/19/25, 5:00PM
27	Sprint Retrospective	26	1 day	11/20/25, 8:00AM	11/20/25, 5:00PM
28	Backlog Refinement	27	1 day	11/21/25, 8:00AM	11/21/25, 5:00PM
29	**Sprint 2: Digital Health Passport**		**20 days**	**11/24/25, 8:00AM**	**12/19/25, 5:00PM**
30	Sprint Planning	28	1 day	11/24/25, 8:00AM	11/24/25, 5:00PM
31	Develop Add Pet Profile Page (UI, Frontend, Backend, APIs,...	30	16 days	11/25/25, 8:00AM	12/16/25, 5:00PM
32	Develop Manage Health Record Page (UI, Frontend, Backen...	30	16 days	11/25/25, 8:00AM	12/16/25, 5:00PM
33	Sprint Review	32	1 day	12/17/25, 8:00AM	12/17/25, 5:00PM
34	Sprint Retrospective	33	1 day	12/18/25, 8:00AM	12/18/25, 5:00PM
35	Backlog Refinement	34	1 day	12/19/25, 8:00AM	12/19/25, 5:00PM
36	**Sprint 3: Intelligent Reminders**		**20 days**	**12/22/25, 8:00AM**	**1/16/26, 5:00PM**
37	Sprint Planning	35	1 day	12/22/25, 8:00AM	12/22/25, 5:00PM
38	Develop Schedule Page (UI, Frontend, Backend, APIs, Data...	37	16 days	12/23/25, 8:00AM	1/13/26, 5:00PM
39	Develop Manage Reminder Page (UI, Frontend, Backend, A...	37	16 days	12/23/25, 8:00AM	1/13/26, 5:00PM
40	Sprint Review	39	1 day	1/14/26, 8:00AM	1/14/26, 5:00PM
41	Sprint Retrospective	40	1 day	1/15/26, 8:00AM	1/15/26, 5:00PM
42	Backlog Refinement	41	1 day	1/16/26, 8:00AM	1/16/26, 5:00PM
43	**Sprint 4: Safety & Tracking Hub**		**20 days**	**1/19/26, 8:00AM**	**2/13/26, 5:00PM**
44	Sprint Planning	42	1 day	1/19/26, 8:00AM	1/19/26, 5:00PM
45	Develop Mange Safety & Tracking Hub Page (UI, Frontend, ...	44	16 days	1/20/26, 8:00AM	2/10/26, 5:00PM
46	Sprint Review	45	1 day	2/11/26, 8:00AM	2/11/26, 5:00PM
47	Sprint Retrospective	46	1 day	2/12/26, 8:00AM	2/12/26, 5:00PM
48	Backlog Refinement	47	1 day	2/13/26, 8:00AM	2/13/26, 5:00PM
49	**Sprint 5: AI-Powered Nutrition**		**20 days**	**2/16/26, 8:00AM**	**3/13/26, 5:00PM**
50	Sprint Planning	48	1 day	2/16/26, 8:00AM	2/16/26, 5:00PM
51	Develop Manage AI-Powered Nutrition Page (UI, Frontend, ...	50	16 days	2/17/26, 8:00AM	3/10/26, 5:00PM
52	Gathering Nutrition Data (Scraping data and Manage Nutriti...	50	16 days	2/17/26, 8:00AM	3/10/26, 5:00PM
53	Sprint Review	52	1 day	3/11/26, 8:00AM	3/11/26, 5:00PM
54	Sprint Retrospective	53	1 day	3/12/26, 8:00AM	3/12/26, 5:00PM
55	Backlog Refinement	54	1 day	3/13/26, 8:00AM	3/13/26, 5:00PM
56	**Sprint 6: AI-Powered Wellness Dashboard**		**20 days**	**3/16/26, 8:00AM**	**4/10/26, 5:00PM**
57	Sprint Planning	55	1 day	3/16/26, 8:00AM	3/16/26, 5:00PM
58	Develop AI-Powered Wellness Dashboard Page (UI, Fronte...	57	16 days	3/17/26, 8:00AM	4/7/26, 5:00PM
59	Develop Wellness Analytics Models	57	16 days	3/17/26, 8:00AM	4/7/26, 5:00PM
60	Sprint Review	59	1 day	4/8/26, 8:00AM	4/8/26, 5:00PM
61	Sprint Retrospective	60	1 day	4/9/26, 8:00AM	4/9/26, 5:00PM
62	Backlog Refinement	61	1 day	4/10/26, 8:00AM	4/10/26, 5:00PM
63	**Sprint 7: Digital Health Certificate**		**20 days**	**4/13/26, 8:00AM**	**5/8/26, 5:00PM**
64	Sprint Planning	62	1 day	4/13/26, 8:00AM	4/13/26, 5:00PM
65	Develop Manage Digital Health Certificate Page (UI, Fronten...	64	16 days	4/14/26, 8:00AM	5/5/26, 5:00PM
66	Sprint Review	65	1 day	5/6/26, 8:00AM	5/6/26, 5:00PM
67	Sprint Retrospective	66	1 day	5/7/26, 8:00AM	5/7/26, 5:00PM
68	Backlog Refinement	67	1 day	5/8/26, 8:00AM	5/8/26, 5:00PM
69	**Sprint 8: Vet Portal & Clinic integration**		**20 days**	**5/11/26, 8:00AM**	**6/5/26, 5:00PM**
70	Sprint Planning	68	1 day	5/11/26, 8:00AM	5/11/26, 5:00PM
71	Develop Vet Portal (UI, Frontend, Backend, APIs, Databases)	70	16 days	5/12/26, 8:00AM	6/2/26, 5:00PM
72	Sprint Review	71	1 day	6/3/26, 8:00AM	6/3/26, 5:00PM
73	Sprint Retrospective	72	1 day	6/4/26, 8:00AM	6/4/26, 5:00PM
74	Backlog Refinement	73	1 day	6/5/26, 8:00AM	6/5/26, 5:00PM
75	**Sprint 9: Community Marketplace - Loyalty Program**		**20 days**	**6/8/26, 8:00AM**	**7/3/26, 5:00PM**
76	Sprint Planning	74	1 day	6/8/26, 8:00AM	6/8/26, 5:00PM
77	Develop Community Marketplace Page (UI, Frontend, APIs)	76	16 days	6/9/26, 8:00AM	6/30/26, 5:00PM
78	Develop Loyalty Program Page (UI, Frontend, Backend, API...	76	16 days	6/9/26, 8:00AM	6/30/26, 5:00PM
79	Sprint Review	78	1 day	7/1/26, 8:00AM	7/1/26, 5:00PM
80	Sprint Retrospective	79	1 day	7/2/26, 8:00AM	7/2/26, 5:00PM
81	Backlog Refinement	80	1 day	7/3/26, 8:00AM	7/3/26, 5:00PM
82	**Sprint 10: Community Marketplace - Adoption**		**20 days**	**7/6/26, 8:00AM**	**7/31/26, 5:00PM**
83	Sprint Planning	81	1 day	7/6/26, 8:00AM	7/6/26, 5:00PM
84	Develop Adoption Page (UI, Frontend, Backend, APIs, Data...	83	16 days	7/7/26, 8:00AM	7/28/26, 5:00PM
85	Sprint Review	84	1 day	7/29/26, 8:00AM	7/29/26, 5:00PM
86	Sprint Retrospective	85	1 day	7/30/26, 8:00AM	7/30/26, 5:00PM
87	Backlog Refinement	86	1 day	7/31/26, 8:00AM	7/31/26, 5:00PM
88	**Sprint 11: Community Marketplace - Insurance Services**		**20 days**	**8/3/26, 8:00AM**	**8/28/26, 5:00PM**
89	Sprint Planning	87	1 day	8/3/26, 8:00AM	8/3/26, 5:00PM
90	Develop Insurance Services Page (UI, Frontend, Backend, A...	89	16 days	8/4/26, 8:00AM	8/25/26, 5:00PM
91	Sprint Review	90	1 day	8/26/26, 8:00AM	8/26/26, 5:00PM
92	Sprint Retrospective	91	1 day	8/27/26, 8:00AM	8/27/26, 5:00PM
93	Backlog Refinement	92	1 day	8/28/26, 8:00AM	8/28/26, 5:00PM
94	**Sprint 12: Multi language Support**		**20 days**	**8/31/26, 8:00AM**	**9/25/26, 5:00PM**
95	Sprint Planning	93	1 day	8/31/26, 8:00AM	8/31/26, 5:00PM
96	Develop AI Translation Integration	95	16 days	9/1/26, 8:00AM	9/22/26, 5:00PM
97	Develop Locale-Aware UI	95	16 days	9/1/26, 8:00AM	9/22/26, 5:00PM
98	Sprint Review	97	1 day	9/23/26, 8:00AM	9/23/26, 5:00PM
99	Sprint Retrospective	98	1 day	9/24/26, 8:00AM	9/24/26, 5:00PM
100	Backlog Refinement	99	1 day	9/25/26, 8:00AM	9/25/26, 5:00PM
101	**Sprint 13: Third-Party Integrations & APIs**		**20 days**	**9/28/26, 8:00AM**	**10/23/26, 5:00PM**
102	Sprint Planning	100	1 day	9/28/26, 8:00AM	9/28/26, 5:00PM
103	Develop API Gateway & Adapters	102	16 days	9/29/26, 8:00AM	10/20/26, 5:00PM
104	Sprint Review	103	1 day	10/21/26, 8:00AM	10/21/26, 5:00PM
105	Sprint Retrospective	104	1 day	10/22/26, 8:00AM	10/22/26, 5:00PM
106	Backlog Refinement	105	1 day	10/23/26, 8:00AM	10/23/26, 5:00PM
107	**Monitoring and Controlling**		**315 days?**	**9/1/25, 8:00AM**	**11/13/26, 5:00PM**
108	Security Monitoring		315 days?	9/1/25, 8:00AM	11/13/26, 5:00PM
109	Risk Monitoring		315 days?	9/1/25, 8:00AM	11/13/26, 5:00PM
110	Progress Monitoring		315 days?	9/1/25, 8:00AM	11/13/26, 5:00PM
111	**Closing**		**15 days?**	**10/26/26, 8:00AM**	**11/13/26, 5:00PM**
112	Create Project Documents	106	10 days?	10/26/26, 8:00AM	11/6/26, 5:00PM
113	Release Candidate Build	106	5 days?	10/26/26, 8:00AM	10/30/26, 5:00PM
114	Regression & Security Testing	106	5 days?	10/26/26, 8:00AM	10/30/26, 5:00PM
115	Lessons Learned Workshop	112	5 days?	11/9/26, 8:00AM	11/13/26, 5:00PM
116	Project Sign-Off	115	0 days?	11/13/26, 5:00PM	11/13/26, 5:00PM

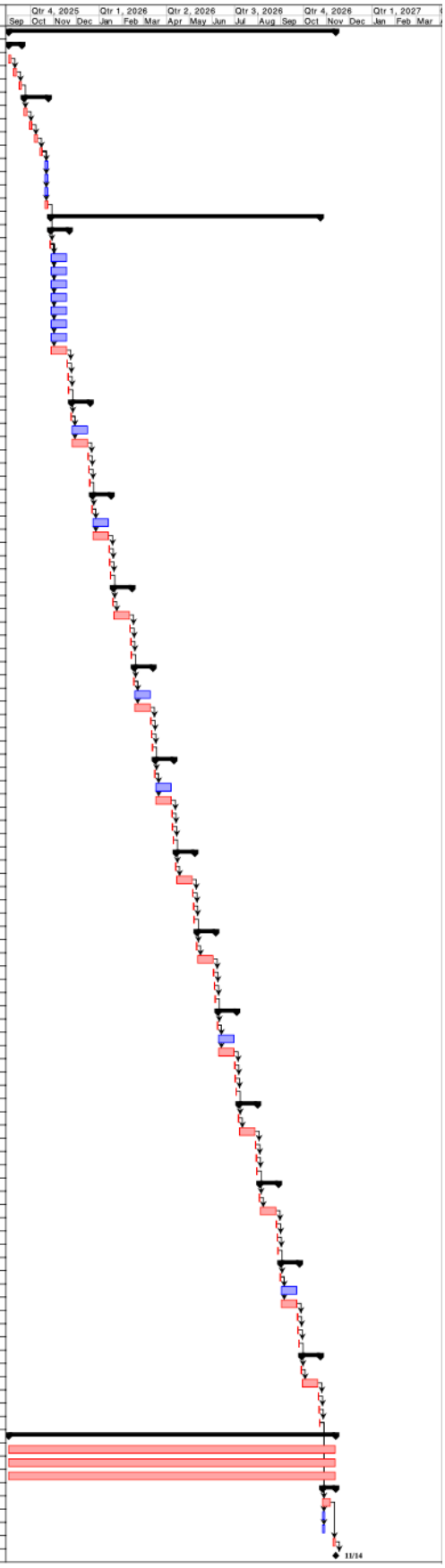


Figure 7 PetCare Gantt chart

6.2. Critical path

From the analysis of the Critical Path in the PetCare Application Project, which follows the Agile Framework, it was found that the sequence of critical activities represents essential tasks that must be executed continuously from project initiation (September 2025) through Phase 2 completion (31 July 2026) for proof-of-concept delivery. Phase 3 (1 Aug 2026 – 13 Nov 2026) execution is conditional on Phase 2 go/no-go decision. Any delay in these activities will directly impact the overall project timeline, potentially extending the completion date and affecting the delivery schedule and development milestones.

1. Project Charter
2. Stakeholder Register
3. Kickoff Meeting
4. Requirements Elicitation
5. Stakeholder Interviews
6. Competitor Research
7. Define Release Plan and Scope of Work
8. UX Wireframes
9. Architecture Baseline
10. Test & Security Planning
11. Tooling Setup (CI/CD, Jira, GitHub)
12. Sprint 1
13. Sprint 2
14. Sprint 3
15. Sprint 4
16. Sprint 5
17. Sprint 6
18. Sprint 7
19. Sprint 8
20. Sprint 9
21. Sprint 10
22. Sprint 11
23. Sprint 12
24. Sprint 13

25. Create Project Documents
26. Release Candidate Build
27. Regression & Security Testing
28. Lessons Learned Workshop
29. Project Sign-Off

7. Resource rate

Table 29 PetCare Resource rate

No.	Resource Name	Cost (THB)	Unit
1	CTO/Product Owner (1)	120,000	Per Month
2	Project Manager (1)	70,000	Per Month
3	Technical Lead (1)	70,000	Per Month
4	Senior Business Analyst (1)	50,000	Per Month
5	Senior System Analyst (1)	50,000	Per Month
6	Senior UX/UI Designer (1)	40,000	Per Month
7	Senior Fullstack Developer (1)	55,000	Per Month
8	Junior Fullstack Developer (2)	30,000	Per Month
9	Senior Software Quality Assurance Engineer (1)	40,000	Per Month
10	Senior DevOps Engineer (1)	60,000	Per Month
11	Security Specialist (1) -> Calculate by job	10,000	Per Day (7 Days/Time/Year)
12	Data Scientists & AI Engineer (1)	70,000	Per Month
13	Marketing Officer (1)	30,000	Per Month
14	Sales Officer (1)	30,000	Per Month
15	Customer Support (1)	25,000	Per Month
16	Veterinary Advisor (1) -> Calculate by job	3,000	Per Day (10 Days/Time/Year)
17	Microsoft 365 (3 USD)	99	Per User Per Month
18	Github (21 USD)	693	Per User Per Month
19	Docker (15 USD)	495	Per User Per Month
20	Azure AI Vision (0.4 USD)	13.2	Per Transaction
21	Azure Kubernetes Service (0.1 USD) (3 Cluster)	3.3	Per Cluster Per Hour
22	Domain (1st Year 225.73/Year)	698.57	698.57 Per Year
23	Azure Monitoring (0.65 USD)	21.45	Per GB
24	Azure API Management (150.01 USD)	4,950.33	Per Month
25	Azure Key Vault – Secret Ops (0.03 USD)	0.99	Per 10,000 Transactions

26	Azure Blob Storage (0.02 USD)	0.66	Per GB

**** 1 USD = 33 THB**

8. Resource assignment to tasks

Table 30 PetCare Resource assignment to tasks

PetCare Mobile Application	Resource Assignment
1. Initiation	
1.1. Project Charter	Product Owner/CTO, Project Manager
1.2. Stakeholder Register	Project Manager, Business Analyst
1.3. Kickoff Meeting	Product Owner/CTO, Project Manager, Technical Lead, Business Analyst, System Analyst
2. Planning	
2.1. Requirements Elicitation	Business Analyst
2.2. Stakeholder Interviews	Business Analyst, Sales Officer
2.3. Competitor Research	Business Analyst, Marketing Officer, Ux/Ui Designer
2.4. Define Release Plan and Scope of Work	Product Owner/CTO, Project Manager, Tech Lead
2.5. UX Wireframes	Ux/Ui Designer
2.6. Architecture Baseline	Technical Lead, System Analyst
2.7. Test & Security Planning	Software QA Engineer, Security Specialist
2.8. Tooling Setup (CI/CD, Jira, GitHub)	DevOps Engineer
3. Executing	
3.1. Sprint 1: Onboarding	
3.1.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.1.2. Develop Account Registration Page (UI, Frontend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.1.3. Develop Log in Page (UI, Frontend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.1.4. Develop Notifications (UI, Frontend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.1.5. Develop Onboarding Page (UI, Frontend)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.1.6. Develop Setting Page (UI, Frontend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer

3.1.7. Develop AI Assistant Page (UI, Frontend, Backend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Data Scientist/AI Engineer
3.1.8. Develop Payment Gateway (APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.1.9. Develop Landing Page (UI, Frontend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.1.10. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer
3.1.11. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, Data Scientist/AI Engineer
3.1.12. Backlog Refinement	Project Manager, Business Analyst
3.2. Sprint 2: Digital Health Passport	
3.2.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.2.2. Develop Add Pet Profile Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.2.3. Develop Manage Health Record Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.2.4. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer, Veterinary Advisor
3.2.5. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst
3.2.6. Backlog Refinement	Project Manager, Business Analyst
3.3. Sprint 3: Intelligent Reminders	
3.3.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.3.2. Develop Schedule Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer

3.3.3. Develop Manage Reminder Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.3.4. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer
3.3.5. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst
3.3.6. Backlog Refinement	Project Manager, Business Analyst
3.4. Sprint 4: Safety & Tracking Hub	
3.4.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.4.2. Develop Manage Safety & Tracking Hub Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.4.3. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer
3.4.4. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst
3.4.5. Backlog Refinement	Project Manager, Business Analyst
3.5. Sprint 5: AI-Powered Nutrition	
3.5.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.5.2. Develop Manage AI-Powered Nutrition Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.5.3. Gathering Nutrition Data (Scraping data and Manage Nutrition DB)	Data Scientist/AI Engineer
3.5.4. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead,

	Business Analyst, System Analyst, Data Scientist/AI Engineer, Veterinary Advisor
3.5.5. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, Data Scientist/AI Engineer
3.5.6. Backlog Refinement	Project Manager, Business Analyst
3.6. Sprint 6: AI-Powered Wellness Dashboard	
3.6.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.6.2. Develop Manage AI-Powered Nutrition Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.6.3. Gathering Nutrition Data (Scraping data and Manage Nutrition DB)	Data Scientist/AI Engineer
3.6.4. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer, Veterinary Advisor
3.6.5. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, Data Scientist/AI Engineer
3.6.6. Backlog Refinement	Project Manager, Business Analyst
3.7. Sprint 7: Digital Health Certificate	
3.7.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.7.2. Develop Manage Digital Health Certificate Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.7.3. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer
3.7.4. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, Veterinary Advisor

3.7.5. Backlog Refinement	Project Manager, Business Analyst
3.8. Sprint 8: Vet Portal & Clinic integration	
3.8.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.8.2. Develop Manage Digital Health Certificate Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.8.3. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst
3.8.4. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, Data Scientist/AI Engineer, Veterinary Advisor
3.8.5. Backlog Refinement	Project Manager, Business Analyst
3.9. Sprint 9: Community Marketplace - Loyalty Program	
3.9.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.9.2. Develop Community Marketplace Page (UI, Frontend, APIs)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.9.3. Develop Loyalty Program Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.9.4. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer
3.9.5. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst
3.9.6. Backlog Refinement	Project Manager, Business Analyst
3.10. Sprint 10: Community Marketplace - Adoption	
3.10.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.10.2. Develop Adoption Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer

3.10.3. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer
3.10.4. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst
3.10.5. Backlog Refinement	Project Manager, Business Analyst
3.11. Sprint 11: Community Marketplace - Insurance Services	
3.11.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.11.2. Develop Adoption Page (UI, Frontend, Backend, APIs, Databases)	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.11.3. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer
3.11.4. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst
3.11.5. Backlog Refinement	Project Manager, Business Analyst
3.12. Sprint 12: Multi language Support	
3.12.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.12.2. Develop AI Translation Integration	Fullstack Developer, Software QA Engineer, DevOps Engineer, Data Scientist/AI Engineer
3.12.3. Develop Locale-Aware UI	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer
3.12.4. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer
3.12.5. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, Data Scientist/AI Engineer
3.12.6. Backlog Refinement	Project Manager, Business Analyst

3.13. Sprint 13: Third-Party Integrations & APIs	
3.13.1. Sprint Planning	Project Manager, Business Analyst, Technical Lead
3.13.2. Develop API Gateway & Adapters	Fullstack Developer, Software QA Engineer, DevOps Engineer
3.13.3. Sprint Review	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst, System Analyst, Data Scientist/AI Engineer
3.13.4. Sprint Retrospective	UX/UI Designer, Fullstack Developer, Software QA Engineer, DevOps Engineer, Technical Lead, Business Analyst
3.13.5. Backlog Refinement	Project Manager, Business Analyst
4. Monitoring and Controlling	
4.1. Security Monitoring	Security Specialist, DevOps Engineer
4.2. Risk Monitoring	Project Manager
4.3. Progress Monitoring	Project Manager
5. Closing	
5.1. Create Project Documents	Business Analyst, System Analyst, Software QA Engineer
5.2. Release Candidate Build	DevOps Engineer
5.3. Regression & Security Testing	Software QA Engineer, Security Specialist
5.4. Lessons Learned Workshop	Product Owner/CTO, Project Manager, Technical Lead, Business Analyst, System Analyst
5.5. Project Sign-Off	Product Owner/CTO, Project Manager

9. Cost baseline

Project Name: The PetCare Mobile Application

Project Start Date: 1 September 2025

Project Finish Date: 13 November 2026

Schedule Estimation: 292 days (Excluding holiday and weekend)

Table 31 Expense table for year 0 between Sep – Dec (Staff Cost)

Operation Expenses (OPEX)	Head Counts	Remuneration	Year 0 (Sep)	Year 0 (Oct)	Year 0 (Nov)	Yeao 0 (Dec)	Total
Staff Cost							
CTO/Product Owner	1	120,000	120,000	120,000	120,000	120,000	480,000
Project Manager	1	70,000	70,000	70,000	70,000	70,000	280,000
Technical Lead	1	70,000	70,000	70,000	70,000	70,000	280,000
Senior Business Analyst	1	50,000	50,000	50,000	50,000	50,000	200,000
Senior System Analyst	1	50,000	50,000	50,000	50,000	50,000	200,000
Senior UX/UI Designer	1	40,000	40,000	40,000	40,000	40,000	160,000
Senior Fullstack Developer	1	55,000	55,000	55,000	55,000	55,000	220,000
Junior Fullstack Developer	2	30,000	60,000	60,000	60,000	60,000	240,000
Senior Software Quality Assurance Engineer	1	40,000	40,000	40,000	40,000	40,000	160,000
Senior DevOps Engineer	1	60,000	60,000	60,000	60,000	60,000	240,000
Security Specialist (Outsource)	1	0	0	0	0	0	0
Data Scientist & AI Engineer	1	70,000	70,000	70,000	70,000	70,000	280,000
Marketing Officer	1	30,000	30,000	30,000	30,000	30,000	120,000
Sales Officer	1	30,000	30,000	30,000	30,000	30,000	120,000
Customer Support	1	25,000	25,000	25,000	25,000	25,000	100,000
Veterinary Advisor	1	0	0	0	0	0	0
Total Staff Cost	17	740,000	770,000	770,000	770,000	770,000	3,080,000

Table 32 Expense table for year 0 between Sep – Dec (Software Cost)

Operation Expenses (OPEX)	Units	Subscription Cost	Year 0 (Sep)	Year 0 (Oct)	Year 0 (Nov)	Yeao 0 (Dec)	Total
Software Cost							
Microsoft 365	15	99	1,485	1,485	1,485	1,485	5,940
GitHub Team/Pro	12	693	8,316	8,316	8,316	8,316	33,264
Docker Subscription	6	495	2,970	2,970	2,970	2,970	11,880
Azure AI Vision	1	4,950	4,950	4,950	4,950	4,950	19,800
Azure Kubernetes Service	3	238	713	713	713	713	2,852
Azure Monitoring	50	21	1,073	1,073	1,073	1,073	4,292
Azure API Management	500	1	330	330	330	330	1,320
Azure Key Vault – Secret Ops	1	10	10	10	10	10	40
Azure Blob Storage	1	6,600	6,600	6,600	6,600	6,600	26,400
Domain Name Registration	1	58	58	58	58	58	232
Total Software Cost	590	13,165	26,505	26,505	26,505	26,505	106,020

(Unit: THB)

Table 33 Expense table for development (Year 0-1)

Operation Expenses (OPEX)	Head Counts/ Units	Year 0	Year 1	Total
CTO/Product Owner	1	480,000	1,440,000	1,920,000
Management Team	1	280,000	840,000	1,120,000
Development and Innovation Team	11	1,980,000	5,940,000	7,920,000
Marketing Team	1	120,000	360,000	480,000
Sales Team	1	120,000	360,000	480,000
Operation Team	1	0	300,000	300,000
Consultant	1	0	21,000	21,000
Software Tools	-	106,019	318,061	424,080
Total	-	3,086,019	9,579,061	12,665,080

(Unit: THB)

The number of working days used for system development is 292 days. The budget used during the system development period (Year 0 to Year 1) is 12,665,080 baht.

10. Tool for quality control

To ensure effective quality control within the PetCare mobile application project, a **Fishbone Diagram** was utilized to analyze the root causes behind the issue: "**New feature testing is slow due to heavy manual QA effort**." This tool supports systematic problem analysis by visualizing the contributing factors that lead to the main problem, allowing the team to clearly identify improvement opportunities. The Fishbone Diagram used in this project is structured based on the 6M Framework:

- Manpower (People)
- Methods (Processes)
- Machines (Tools/Infrastructure
- Materials (Test Data/Resources)
- Measurements (Metrics)
- Mother-nature (External Constraints/Environment)

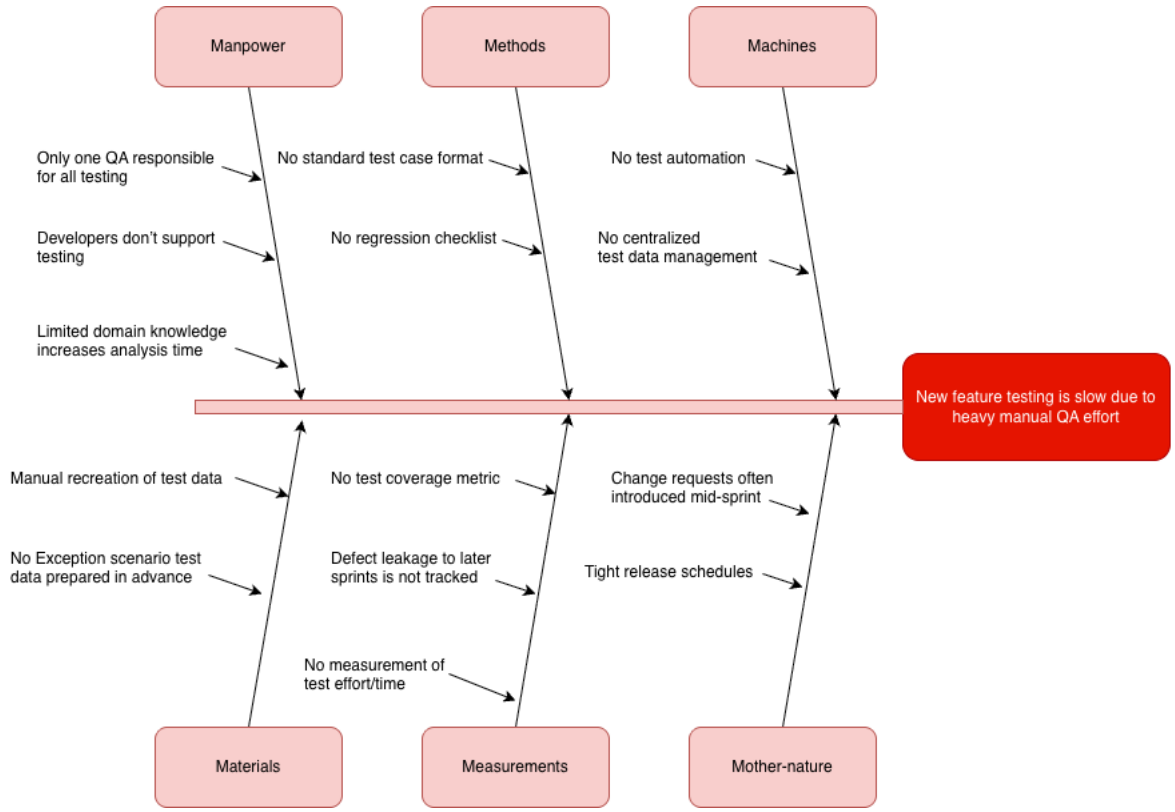


Figure 8 Fish bone diagram

11. Risk management

11.1. Risk register

Table 34 PetCare Risk register (Negative Risk)

ID	Rank	Risk	Description	Category	Root Cause	Triggers	Potential Responses	Risk Owner	Probability	Impact	Status
R1	1	Data Breach/ Unauthorized Access	Sensitive pet and user information may be accessed without permission	Security Risk	Insufficient authentication & access control	Suspicious login activities or failed login attempts	Strengthen access control, enforce MFA, encrypt sensitive data	System Analyst, Security Specialist	Medium	High	Mitigation in progress
R2	2	High Development Cost	Application features require high maintenance & infrastructure costs	Financial Risk	Complex feature set and cloud resources scaling	Rapid increase in monthly cloud cost reports	Optimize resource usage, perform cloud cost monitoring & refactoring	Project Manager	Medium	High	Requires Budget Review
R3	3	System Performance Issues	Application may become slow under heavy usage	Operational Risk	Limited performance testing & optimization	Spike in traffic or crash logs increase	Conduct load testing, implement caching &	Technical Lead, DevOps Engineer	Medium	High	Performance Testing Planned

							optimize DB queries				
R4	4	Third-Party API Dependency Failure	External vet clinic or payment APIs may fail or become unavailable	Integration Risk	Heavy reliance on third-party services	API timeout or service outage	Implement fallback handling, local caching, and retry mechanism	Fullstack Developer, DevOps Engineer	Medium	Medium	Mitigation Strategy Defined

Table 35 PetCare Risk register (Positive Risk)

ID	Rank	Risk	Description	Category	Root Cause	Triggers	Potential Responses	Risk Owner	Probability	Impact	Status
R5	1	Increase in New Users	The app gains more new users than expected due to strong product-market fit	Market Risk	Features align well with pet owners' daily needs	Surge in app sign-ups and downloads	Scale server infrastructure and enhance onboarding flow to retain users	Product Owner/ Technical Lead	High	High	Scaling plan in progress
R6	2	Higher Premium Subscription Rate	A significant portion of users choose to upgrade to premium wellness features	Financial Risk	Increased adoption of AI-powered services within the app	Increase in premium conversions	Introduce additional loyalty tiers and long-term subscription discounts	Product Owner/ Marketing Officer	Medium	High	Premium feature roadmap extended
R7	3	Active User Community Engagement	Users actively participate	Operational Risk	Community features encourage	Increase in forum/group activities	Launch community events,	Marketing Officer	Medium	Medium	Community engagement

			in community features (pet stories, tips, adoption posts)slow under heavy usage		interaction and sharing		contests, or ambassador program				strategy planned

Table 36 PetCare Risk response strategy

ID	Name	Risk Type	Risk Category	Root Cause	Response Strategy
R1	Data Breach/ Unauthorized Access	Negative	Security	Weak access control, insufficient MFA	**Risk Mitigation** – Strengthen access control, enforce MFA, encrypt sensitive data; implement PDPA compliance monitoring.
R2	High Development Cost	Negative	Financial	Complex feature set and cloud scalability	**Risk Mitigation** – Optimize resource usage, apply cost monitoring, and refactor modules to reduce infrastructure load.
R3	System Performance Issues	Negative	Operational	Limited performance testing and DB optimization	**Risk Mitigation** – Conduct load testing, implement caching, tune DB; include this in QA plan.
R4	Third-Party API Dependency Failure	Negative	Integration	Dependence on external APIs and	**Risk Mitigation** – Use fallback handling, retry

				payment gateways	mechanisms, and maintain multi-vendor options.
R5	Increase in New Users	Positive	Market	High product-market fit, effective features	**Risk Exploitation** – Scale infrastructure, enhance onboarding, and maintain retention campaigns.
R6	Higher Premium Subscription Rate	Positive	Financial	Strong interest in AI-driven services	**Risk Enhancement** – Add loyalty tiers, subscription discounts, and AI value upgrades.
R7	Active User Community Engagement	Positive	Operational	Increased participation in community features	**Risk Enhancement** – Host events, contests, and ambassador programs to sustain engagement.

11.2. Risk management matrix and risk evaluate

Table 37 Risk matrix

RISK MATRIX					
		SECTION IMPACTED			
		LOW	MEDIUM	HIGH	EXTREME
LIKELIHOOD	LOW	N	D	C	A
	MEDIUM	D	C	B	A
	HIGH	C	B	A	A

Table 38 Risk Mitigation based upon grade

Grade	Possible Action
A	As a priority, mitigation actions reducing both likelihood and seriousness are to be identified and implemented at the start of project.
B	Mitigation actions reducing both likelihood and seriousness are to be identified and implemented throughout course of project.
C	Mitigation actions reducing both likelihood and seriousness are to be identified and costed for possible action should funds permit execution.
D	Risk to be noted with action only required should grading increase over time.
N	Risk to be noted with action only required should grading increase over time

Table 39 Risk Analysis and Evaluation

Risk Analysis and Evaluation				
No.	IDENTIFIED RISK	RISK GRADE	POTENTIAL IMPACT	RATING KEY
1	Data breach or unauthorized access to pet/user data	High	High	A
2	High development and cloud infrastructure cost	Medium	High	B
3	System performance degradation under load	Medium	High	B
4	Third-party API or integration failure (e.g., Payment Gateway, GPS Tracker)	Medium	Medium	C
5	Increasing new users beyond projected system capacity (Positive Risk)	High	High	B

6	Growing premium subscription adoption (Positive Risk)	Medium	High	**A**
7	Active community engagement and brand advocacy (Positive Risk)	Medium	Medium	**B**

Appendix A

Summary of Revenue Assumptions

Year	Standard	Premium	Breeder Users	Clinic	User per Month	Total Monthly Revenue (THB)	Total annual income (THB)	Percentage of User out of total Pet Parents (7.9 Millions)	Percentage of User out of total Clinic (3,000 Clinics)
2027	2,871	429	10	23	3,300	407090	4885080	0.04%	0.77%
2028	4,308	644	15	32	4,952	607133	7285596	0.06%	1.07%
2029	6,463	967	23	46	7,430	908147	10897764	0.09%	1.53%
2030	9,049	1,355	32	65	10,404	1272564	15270768	0.13%	2.17%
2031	12,676	1,898	45	90	14,574	1781081	21372972	0.19%	3.00%
2032	17,762	2,664	63	125	20,426	2494911	29938932	0.26%	4.17%
2033	24,064	3,610	86	174	27,674	3387440	40649280	0.35%	5.80%
2034	32,487	4,874	116	242	37,361	4583823	55005876	0.47%	8.07%
2035	43,858	6,572	157	337	50,430	6202213	74426556	0.64%	11.23%

Pet parent market size reference:

https://stepplustraining.com/%E0%B8%82%E0%B8%B6%E0%B9%89%E0%B8%99%E0%B8%97%E0%B8%B0%E0%B9%80%E0%B8%9A%E0%B8%B5%E0%B8%A2%E0%B8%99-%E0%B8%AB%E0%B8%A1%E0%B8%B2-%E0%B9%81%E0%B8%A1%E0%B8%A7/?utm_source=chatgpt.com

Total number of clinics reference:

https://www.kachavet.com/%E0%B8%A3%E0%B8%B2%E0%B8%A2%E0%B8%A5%E0%B8%B0%E0%B9%80%E0%B8%AD%E0%B8%B5%E0%B8%A2%E0%B8%94/%E0%B9%82%E0%B8%A3%E0%B8%87%E0%B8%9E%E0%B8%A2%E0%B8%B2%E0%B8%9A%E0%B8%B2%E0%B8%A5%E0%B8%AA%E0%B8%B1%E0%B8%95%E0%B8%A7%E0%B9%8C%E0%B8%84%E0%B8%8A%E0%B8%B2%E0%B9%80%E0%B8%A7%E0%B9%87%E0%B8%97%E0%B8%9C%E0%B9%88%E0%B8%B2%E0%B8%99%E0%B8%A1%E0%B8%B2%E0%B8%95%E0%B8%A3%E0%B8%B2%E0%B8%90%E0%B8%B2%E0%B8%99_Und_TAHSA_Und_%E0%B8%A5%E0%B8%B3%E0%B8%94%E0%B8%B1%E0%B8%9A%E0%B8%97%E0%B8%B5%E0%B9%88_Und_10)

Appendix B

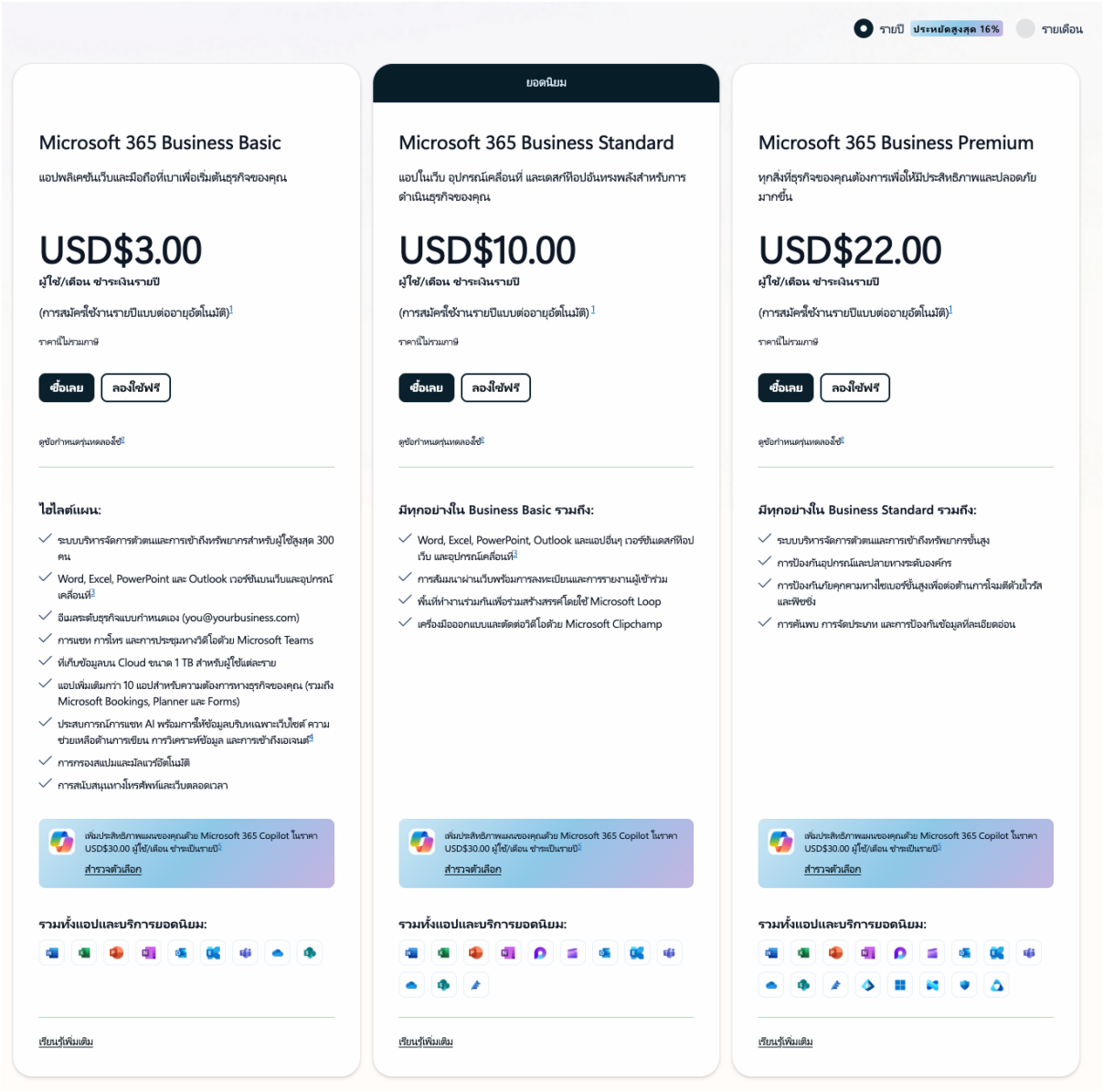
Petcare Software tools list

Software Tools		
Document and Planning tools		
Microsoft 365	Paid	All = 15 Users
Figma	Free	Ux/Ui Designer (1) = 1 User
Draw.io	Free	BA (1), SA (1) = 2 Users
Google Sheet	Free	All = 15 Users
Project Libre	Free	All = 15 Users
Development tools		
VSCode (Vue.JS + Node.JS)	Free	Tech lead (1), Dev (3) = 4 Users
MySQL	Free	Tech lead (1), Dev (3) = 4 Users
Redis	Free	Tech lead (1), Dev (3) = 4 Users
Kafka	Free	Tech lead (1), Dev (3) = 4 Users
Azure AI Vision	Paid	5,000 free transactions per month
Azure API Management	Paid	Subscribe per month
Azure Blob Storage	Paid	~ 100 GB/Month
Testing tools		
Postman	Free	Tech lead (1), Dev (3), QA (1) = 5 Users
JMeter	Free	Tech lead (1), Dev (3) = 4 Users
Robot Framework	Free	Tech lead (1), Dev (3), QA (1) = 5 Users
DevOps tools		
Docker	Paid	DevOps (1), Tech lead (1), Dev (3), AI Engineer (1) = 6 Users
Azure Kubernetes Service	Paid	3 Cluster (Non Prod, Pre Prod, Prod)
Github	Paid	DevOps (1), Tech lead (1), Dev (3), AI Engineer (1) = 6 Users
Jenkins	Free	DevOps (1), Tech lead (1), Dev (3), AI Engineer (1), QA (1) = 7 Users
Deployment tools		
Domain (.com)	Paid	1st Year 225.73/Year 2nd Year onward 698.57/Year
Monitoring tools		
Grafana	Free	DevOps (1), Tech lead (1), Dev (3), AI Engineer (1) = 6 Users

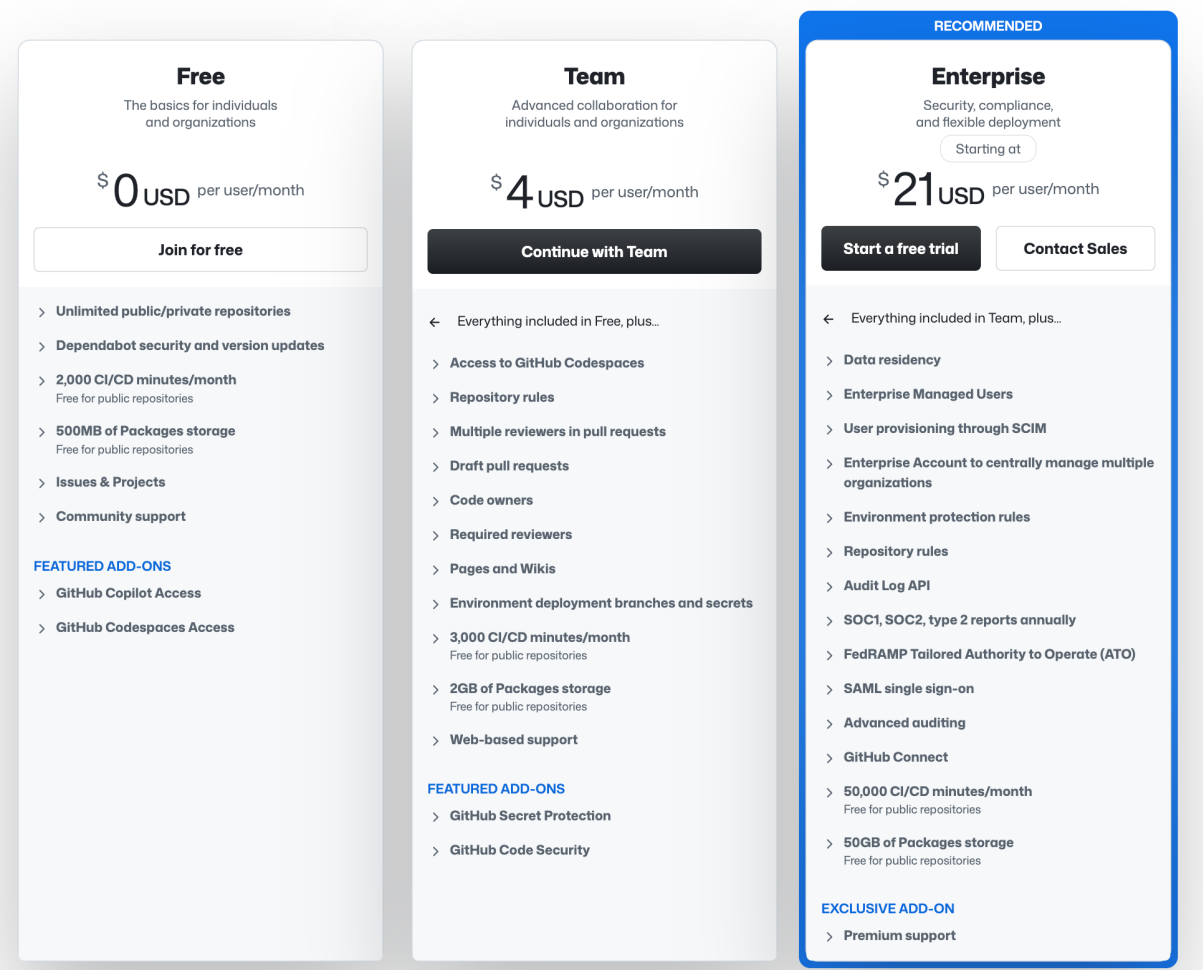
Prometheus	Free	DevOps (1), Tech lead (1), Dev (3), AI Engineer (1) = 6 Users
Azure Monitor	Paid	~ 330 GB/Month
Security tools		
JWT	Free	-
OAuth2	Free	-
TLS	Free	-
Azure Key Vault	Paid	~ 30,000 Transactions/Month

Software Tools Pricing

Microsoft 365 (https://www.microsoft.com/th-th/microsoft-365/business/microsoft-365-plans-and-pricing)



Github action (https://github.com/pricing)



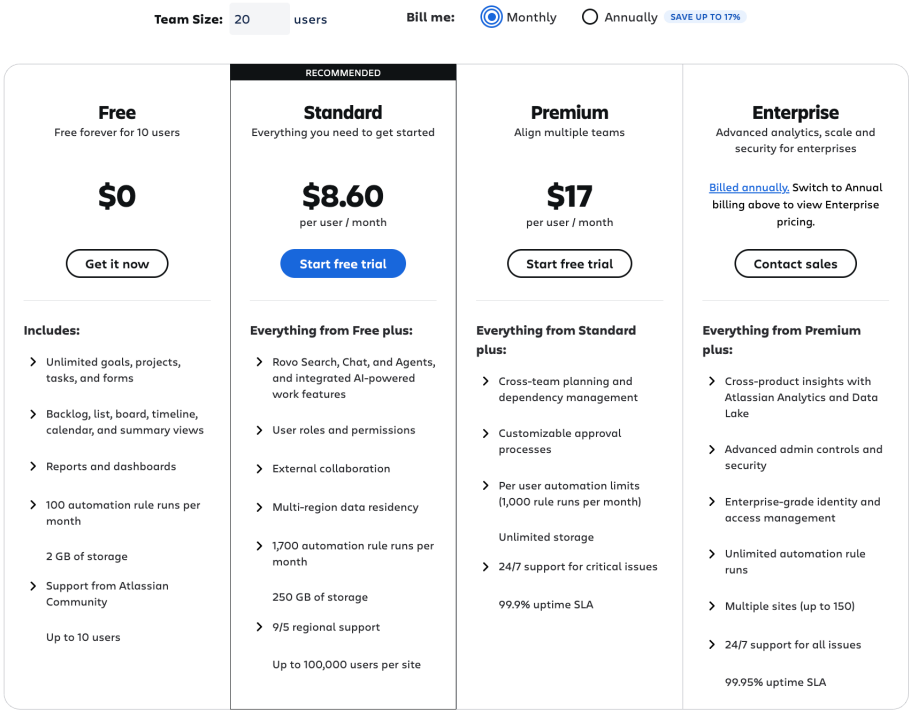
Azure Blob storage (https://azure.microsoft.com/en-us/pricing/details/storage/blobs/)

Data storage prices pay-as-you-go

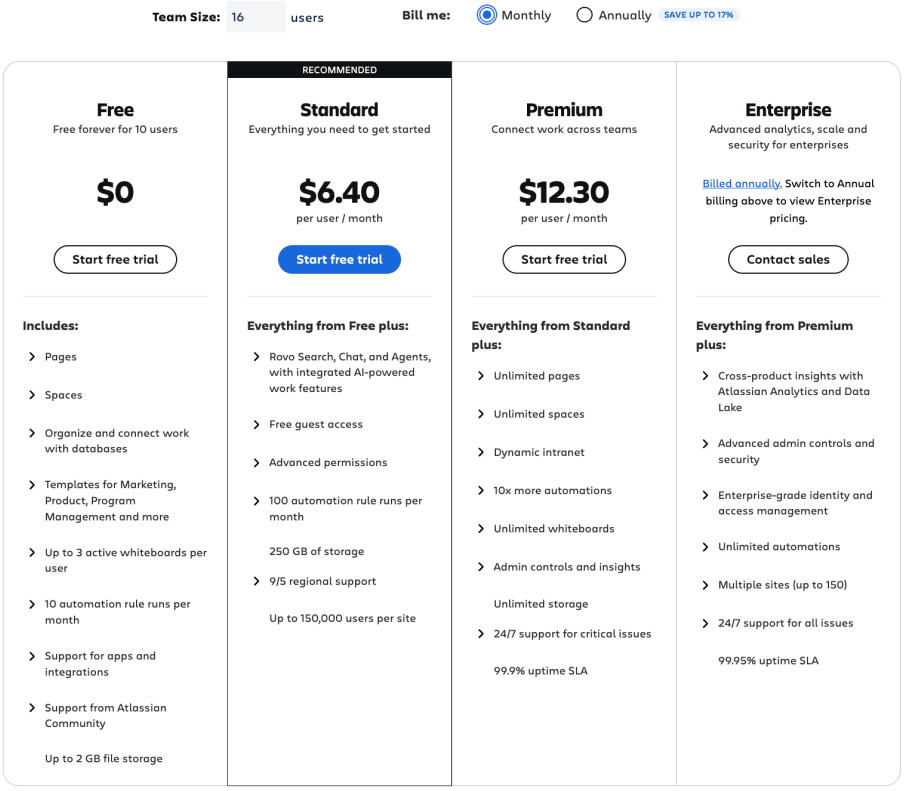
All prices are per GB per month.

Data storage prices pay-as-you-go	Premium	Hot	Cool	Archive
First 50 terabyte (TB)/month	$0.195 per GB	$0.02 per GB	$0.011 per GB	$0.002 per GB
Next 450 TB/month	$0.195 per GB	$0.0192 per GB	$0.011 per GB	$0.002 per GB
Over 500 TB/month	$0.195 per GB	$0.0184 per GB	$0.011 per GB	$0.002 per GB

Jira (https://www.atlassian.com/software/jira/pricing)



Confluence (https://www.atlassian.com/software/confluence/pricing)



Azure AI Vision (https://azure.microsoft.com/en-us/pricing/details/cognitive-services/computer-vision/)

Image Analysis

Instance	Features		Price
Free (F0) - Web/Container	All		5,000 free transactions per month 20 transactions per minute
Standard (S1) - Web/Container	Background Removal (preview)		Free
	Group 1	Tag Face GetThumbnail Color Image Type GetAreaOfInterest People Detection (preview) Smart Crops OCR Adult Celebrity Landmark Object Detection Brand	0-1M transactions - **$1** per 1,000 transactions 1-10M transactions - **$0.65** per 1,000 transactions 10-100M transactions - **$0.60** per 1,000 transactions 100M+ transactions - **$0.40** per 1,000 transactions
	Group 2	Describe Read Caption Dense Captions	0-1M transactions - **$1.50** per 1,000 transactions 1M+ transactions - **$0.60** per 1,000 transactions

Image Analysis Model Customization

Instance	Features	Price
Free (F0) - Web/Container	All	5,000 free inferencing transactions per month 20 transactions per minute
Standard (S1) - Web/Container	Custom Object Detection (preview) Custom Image Classification (preview)	Training: **$20**/Hour Inferencing: **$2**/1K Transactions

Azure API Management (https://azure.microsoft.com/en-us/pricing/details/api-management/#pricing)

API Management (v2 tiers)

	Basic v2	Standard v2	Premium v2 (in preview)
Purpose	API management for teams and projects	Start your organizational API program and scale as it takes off	Operate an enterprise-grade API program
Price[1]	**$150.01** per month	**$700** per month	**$2,801** per month
Scale-out (per additional unit)	**$150.01** per month	**$500** per month	**$1,401** per month
API requests[2] (per month)	10M included **$3** per 1M additional API requests	50M included **$2.50** per 1M additional API requests	Unlimited included
Cache, built-in (per region)	250 MB	1 GB	5 GB
Cache, external[3]	Yes	Yes	Yes
SLA	99.95%	99.95%	99.99%[5]
Multiple custom domain names[4]	No	No	No
Virtual Network	No	VNET integration[6]	VNET integration[6] VNET injection[7]
Self-hosted gateway	No	No	No

Azure Kubernetes Service (https://azure.microsoft.com/en-us/pricing/details/kubernetes-service/#pricing)

AKS pricing tier comparison

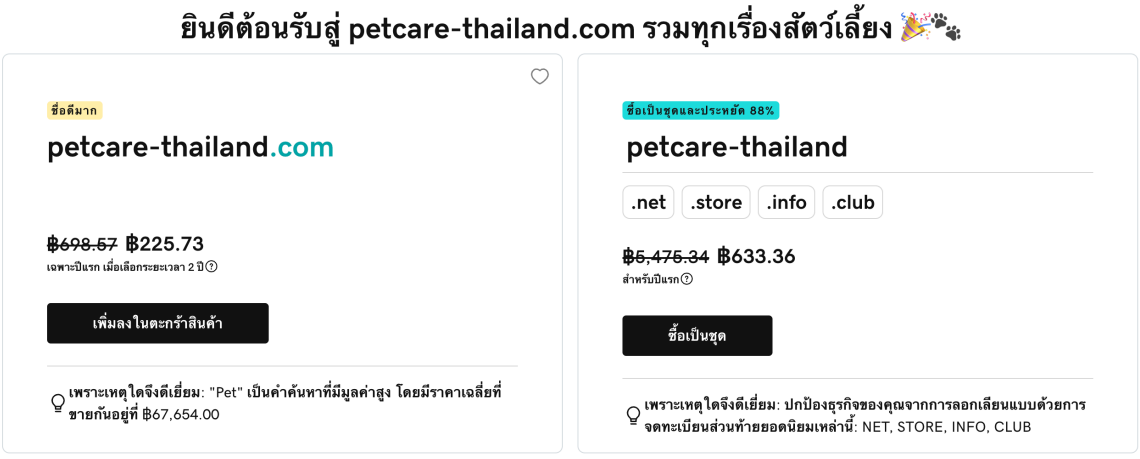
	Azure Kubernetes Service	
	AKS Automatic	**Standard**
Use cases	Production workloads	• Production workloads • Workloads requiring 2 years of support
Node management	Nodes are managed for you	You create and manage nodes in node pools
Cluster node limit	5,000	Up to 5,000*
Service Level Agreement (SLA) options	Financially backed API server uptime SLA	• No SLA (Free tier) • Financially backed API server uptime SLA (Standard tier)
Support options	Community support	Community support with option to purchase Microsoft provided Long Term Support (LTS)**
Pricing	$0.10 per cluster per hour Compute charges are based on the duration and type of virtual machines launched and managed by AKS Automatic.	• No SLA (Free tier): Free (only pay for underlying resources) • SLA (Standard tier): $0.10 per cluster per hour • SLA and LTS (Premium tier): $0.60 per cluster per hour

*Node limit is 1,000 (but not recommended at more than 10 node scale) when operating without an SLA in Free tier.

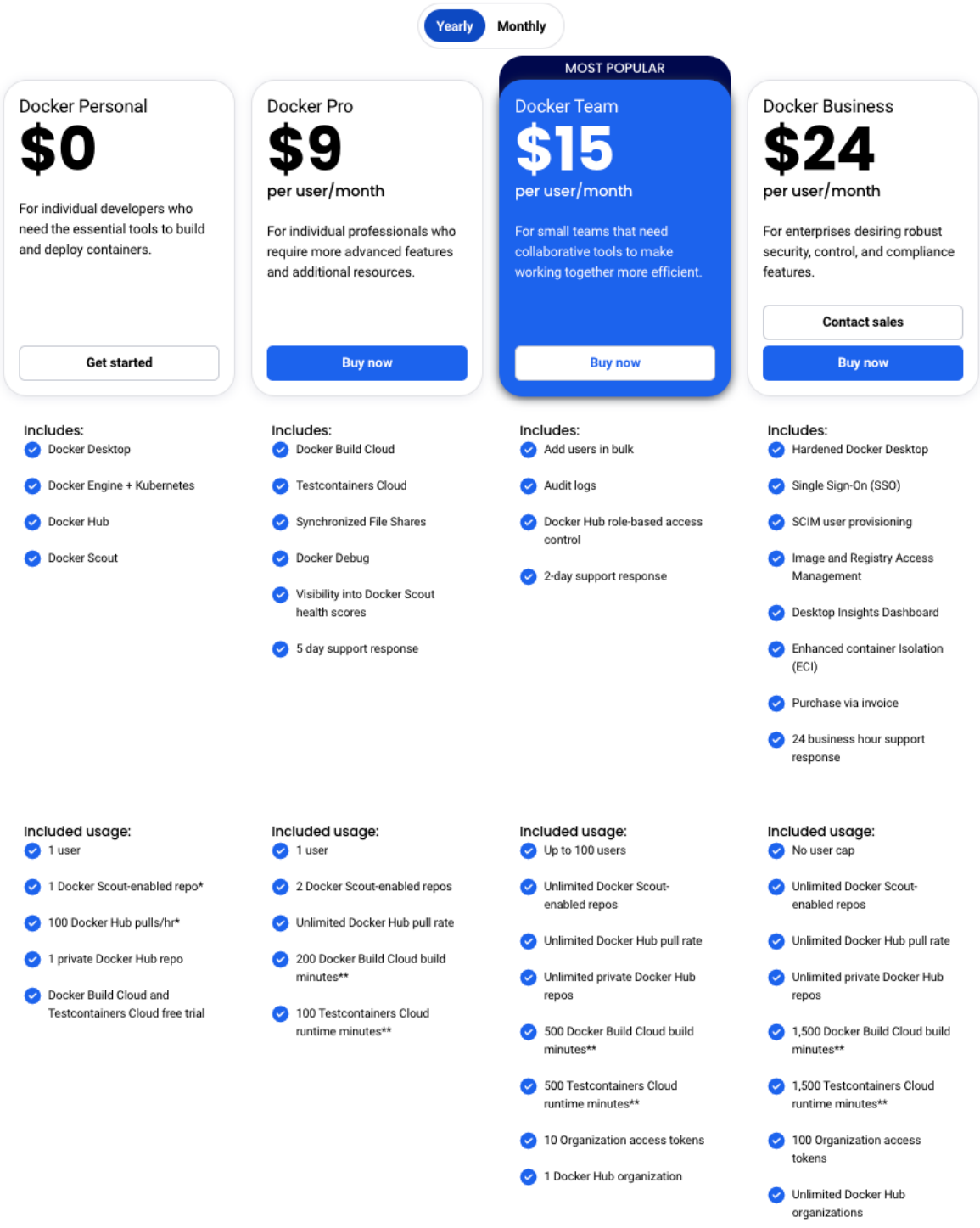
**AKS Long Term Support can be applied to an LTS-capable Kubernetes version to provide extended support from Microsoft for a period of 2 years. LTS is only available for Standard clusters, and clusters operating with LTS will be billed at $0.60 per cluster per hour.

Learn more about the differences between Automatic and Standard.

Domain (https://www.godaddy.com/th-th) (ปีแรก 225.73, ปี 2 fee: 698.57/ปี)



Docker (https://www.docker.com/pricing/)



Azure Monitor (https://azure.microsoft.com/en-us/pricing/details/monitor/)

Ingestion

Azure Monitor Logs charge for data ingested. We offer 3 log ingestion plans – Auxiliary Logs, Basic Logs, and Analytics Logs. A typical customer would use a combination of all plans for different tables in the workspace. For more details on the log plans see here.

Plan	Capabilities	Pricing Tier	Price	Effective Per GB Price[1]	Savings Over Pay-As-You-Go
Auxiliary Logs	Ingestion into the Auxiliary Logs plan: • 30 days of retention included; can be extended up to 12 years • 30 days interactive query; query older data using search jobs • Full KQL on a single table and lookup to Analytics tables • Summary rules	N/A	$0.07 per GB	$0.07 per GB	N/A
	Log processing[2]: • Transforming and filtering data in the Azure Monitor cloud pipeline • All standard log tables include log processing by default.	N/A	$0.13 per GB[3]	$0.13 per GB	N/A
Basic Logs	Ingestion into the Basic Logs plan: • 30 days of retention included; can be extended up to 12 years • 30 days interactive query; query older data using search jobs • Fast query performance with full KQL on a single table and lookup to Analytics tables • Alerting (Simple Log Alerts - Preview) • Summary rules Log processing in the Azure Monitor cloud pipeline is included.	Pay-As-You-Go	$0.65 per GB	$0.65 per GB	N/A
Analytics Logs	• 31/90 days of retention included; can be extended up to 12 years • Up to 2 years interactive query; query older data using search jobs • Query cost included • Fast query performance with full KQL support • Out-of-the-box insights • Alerting • Summary rules	Pay-As-You-Go[4]	$2.99 per GB	$2.99 per GB	N/A
		100 GB per day	$254.80 per day	$2.55 per GB	15%
		200 GB per day	$478.40 per day	$2.40 per GB	20%
		300 GB per day	$702 per day	$2.34 per GB	22%
		400 GB per day	$915.20 per day	$2.29 per GB	23%
		500 GB per day	$1,124.50 per day	$2.25 per GB	25%
		1,000 GB per day	$2,210 per day	$2.21 per GB	26%
		2,000 GB per day	$4,316 per day	$2.16 per GB	28%
		5,000 GB per day	$10,465 per day	$2.10 per GB	30%
		10,000 GB per day	$20,332 per day	$2.04 per GB	32%
		25,000 GB per day	$49,335 per day	$1.98 per GB	34%
		50,000 GB per day	$95,680 per day	$1.92 per GB	36%

[1]Data size is measured in GB (10^9 bytes). Details of its calculation are available for Log Analytics and Application Insights.

[2]Please refer to our documentation for more details and examples of billing scenarios and instances where charges are excluded.

[3]Details on billing start date will be announced on Azure Updates. For current users of the feature, advanced notice will be given before billing starts.

[4]The first 5 GB/month per billing account in this tier are free.

Azure Key Vault (https://azure.microsoft.com/en-us/pricing/details/key-vault/)

Vaults

Vaults are offered in two service tiers—standard and premium.

	Standard	Premium
Secrets operations	**$0.03**/10,000 transactions	**$0.03**/10,000 transactions
Certificate operations[1]	Renewals—**$3** per renewal request All other operations—**$0.03**/10,000 transactions	Renewals—**$3** per renewal request All other operations—**$0.03**/10,000 transactions
Managed Azure Storage account key rotation (in preview)	Free during preview General availability price—**$1** per renewal[2]	Free during preview General availability price—**$1** per renewal[2]

[1]Key Vault does not issue certificates or resell certificates from CAs. Key Vault provides the ability to simplify and automate certain tasks on certificates that you purchase from Public CAs, such as enroll and renew.

[2]Storage account keys are stored as 'secrets' in Key Vault, and therefore operations charges (see 'Secrets Operations' row above) will apply on any operation performed on these keys, including a renewal. See FAQs below for more details on how operations are defined.

Software-protected keys

	Standard	Premium
RSA 2,048-bit keys	**$0.03**/10,000 transactions	**$0.03**/10,000 transactions
Advanced key types— RSA 3,072-bit, RSA 4,096-bit, and Elliptic-Curve Cryptography (ECC) keys	**$0.15**/10,000 transactions	**$0.15**/10,000 transactions

HSM-protected keys

	Standard	Premium
RSA 2,048-bit keys	N/A	**$1** per key per month[1] + **$0.03**/10,000 transactions
Advanced key types[1]— RSA 3,072-bit, RSA 4,096-bit, and Elliptic-Curve Cryptography (ECC) keys	N/A	First 250 keys — **$5** per key per month From 251 – 1,500 keys — **$2.50** per key per month From 1,501 – 4,000 keys — **$0.90** per key per month 4,001+ keys — **$0.40** per key per month + **$0.15**/10,000 transactions

[1]Only actively used HSM protected keys (used in prior 30-day period) are charged, and each version of an HSM protected key is counted as a separate key. See FAQs below for more details. Billing will commence from November 1, 2018.

Key Rotation

	Standard	Premium
Automated key rotation	**$1** per scheduled rotation	**$1** per scheduled rotation

Managed HSM Pools

	Hourly usage fee per HSM pool
Standard B1	$3.20